uc3m
AIRBUS

Airframe Technology Master

Fatigue and Damage Tolerance of metallic and composite structures

Lecture 04
Fatigue endurance.
Stress Concentration Factors

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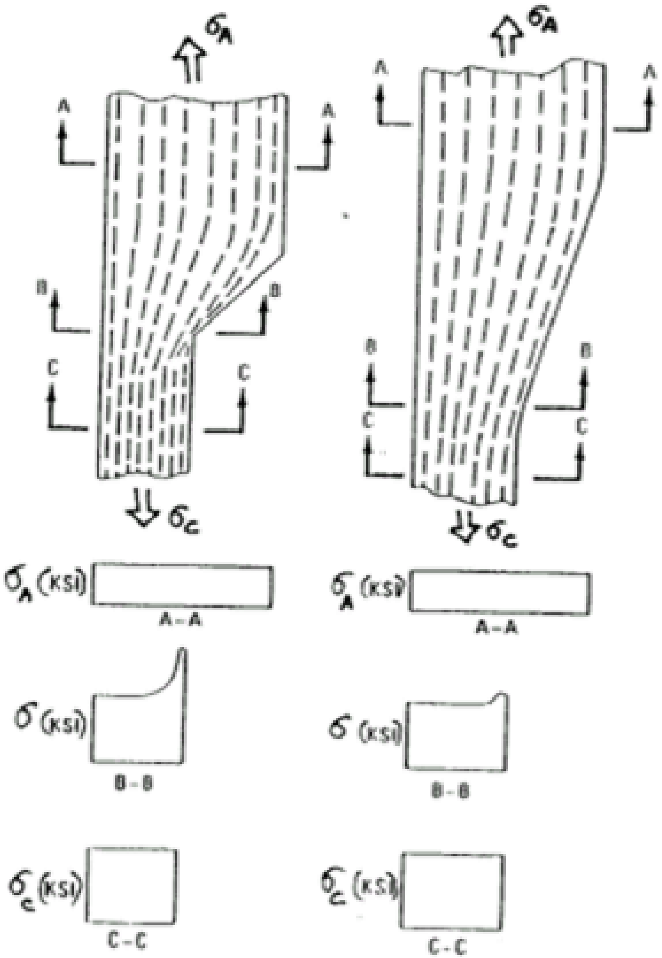
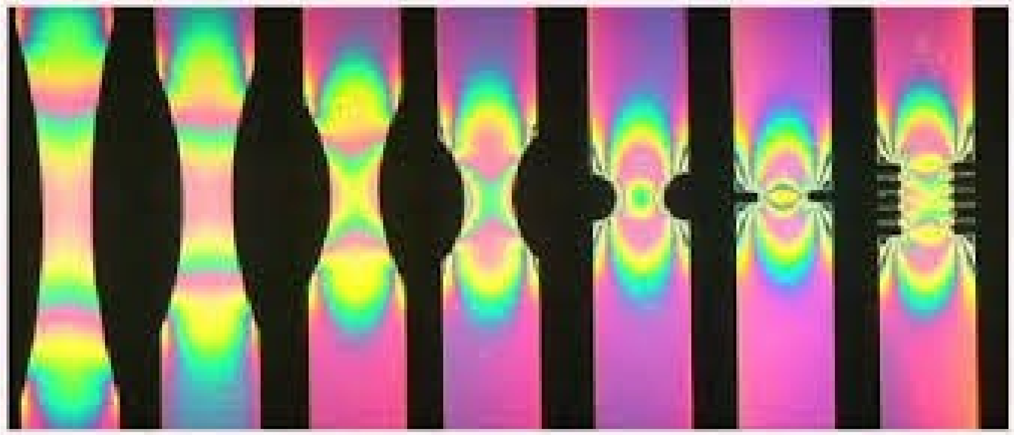
13 October 2025

Contents

04.1 Stress concentration factors

INTRODUCTION





Kt (Stress Concentration Factor)

Relation between:

- Maximum stress at the notch σ
- Reference stress–tension or bending-
  - Real but far away from the critical section (far field stress): Ktgross (Ktg)
  - Virtual – net stress in the section of the critical notch assuming uniform stress : Ktnet
  - Virtual – bearing stress in the section of a loaded hole: Ktbrg

$$Kt = \frac{\sigma_{max}}{\sigma_{ref}}$$

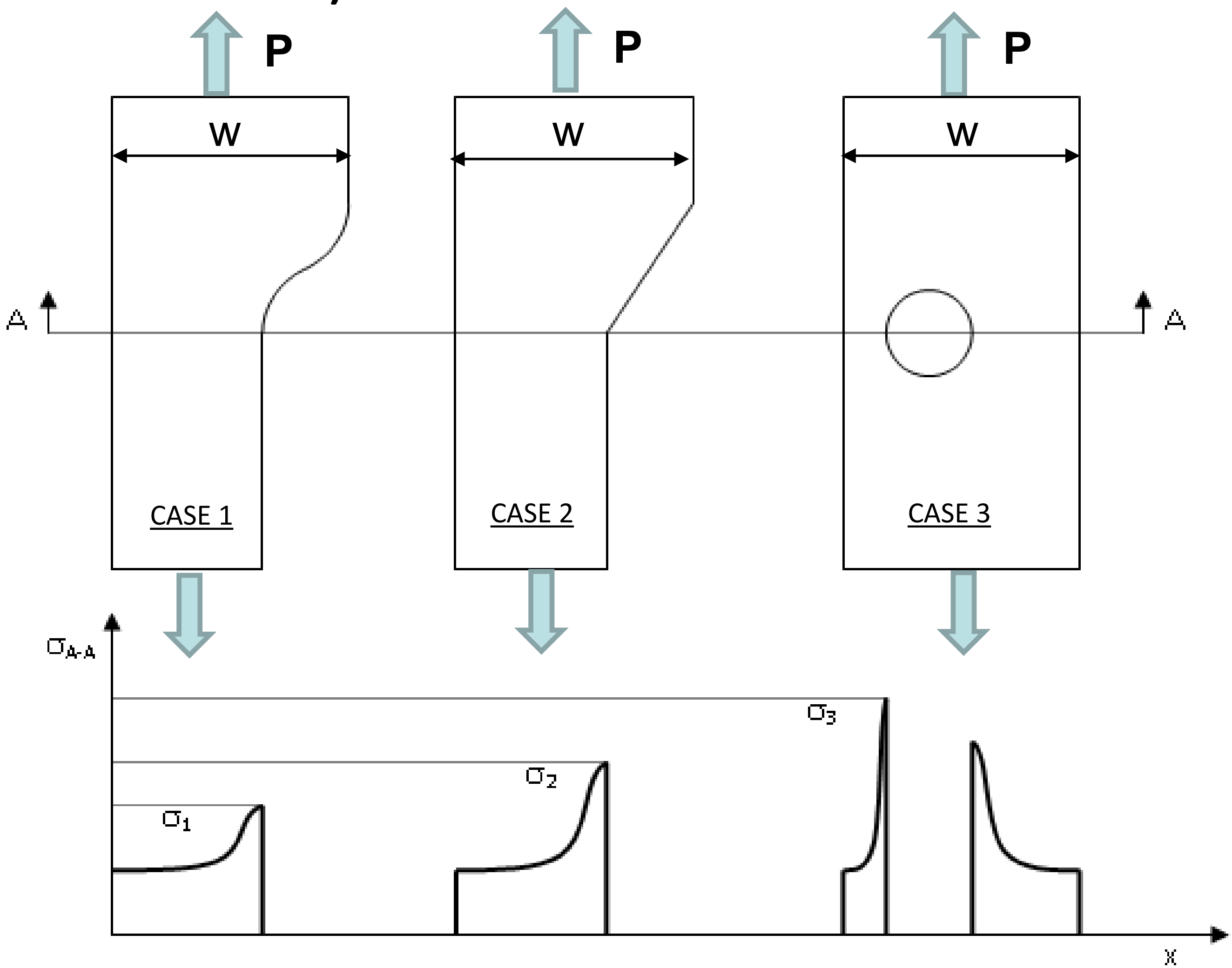
$$\sigma_{ref} = \sigma_{gross} \text{ or } \sigma_{net} \text{ or } \sigma_{brg}$$

STRESS CONCENTRATION FACTORS

Kt (Stress Concentration Factor)

$$Kt_{gross} = \frac{\sigma_{max}}{\sigma_{gross}}$$

$$\sigma_{gross} = P/w \cdot t$$



$$Kt_1 = \frac{\sigma_1}{\sigma_{gross}}$$

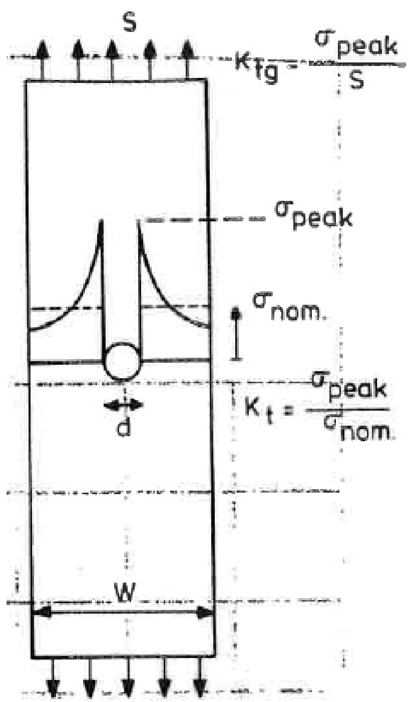
$$Kt_2 = \frac{\sigma_2}{\sigma_{gross}}$$

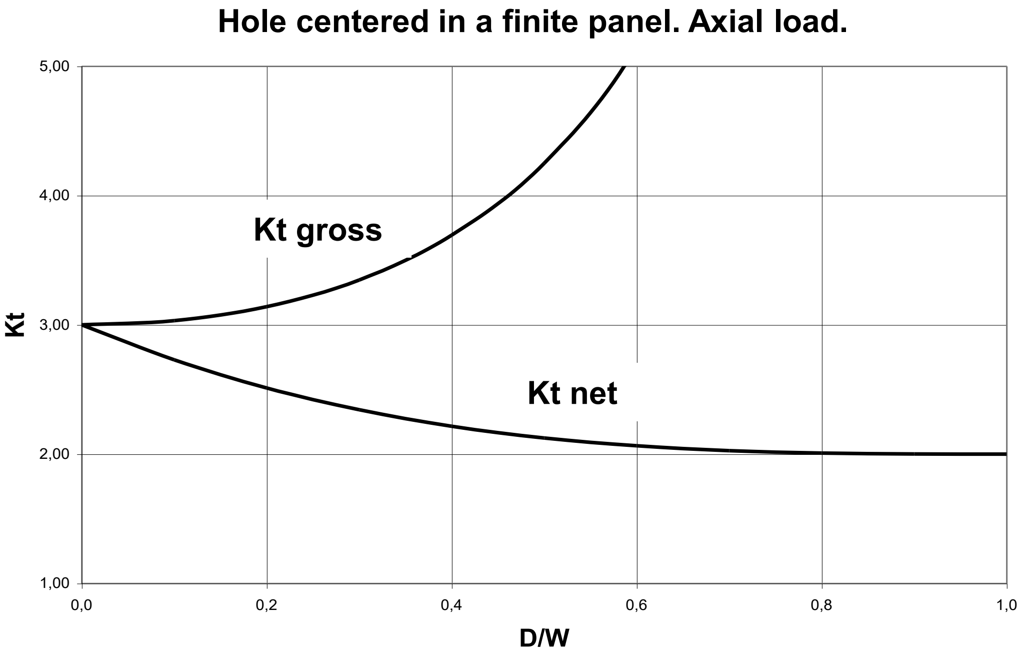
$$Kt_3 = \frac{\sigma_3}{\sigma_{gross}}$$

K_{tGROSS}, K_{tNET} & SAF (Stress Adjustment Factor)

$$Kt_{gross} = \frac{\sigma_{max}}{\sigma_{gross}} \times \frac{\sigma_{net}}{\sigma_{net}} = \frac{\sigma_{max}}{\sigma_{net}} \times \frac{\sigma_{net}}{\sigma_{gross}} = Kt_{net} \times \frac{A_{gross}}{A_{net}} = Kt_{net} \times \frac{1}{(1-d/W)}$$

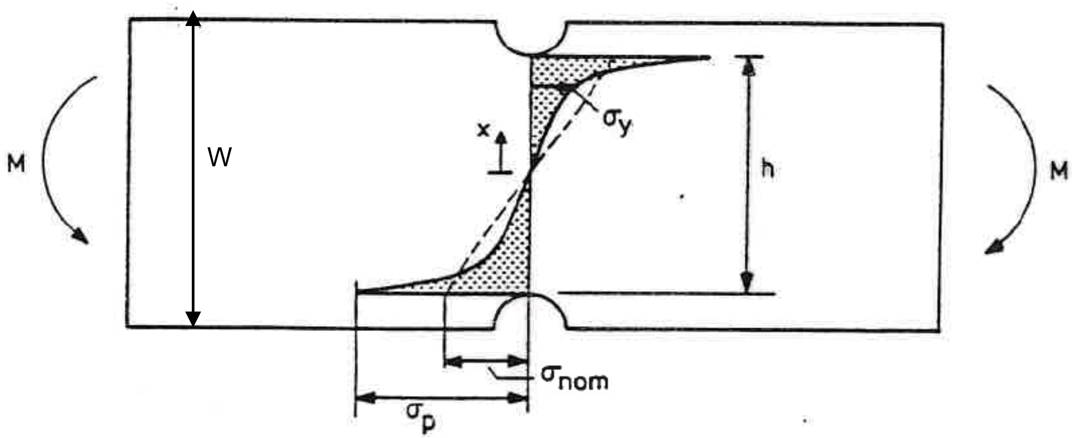
$$SAF_{net} = \frac{Kt_{gross}}{Kt_{net}} = \frac{1}{(1-d/W)}$$





Bibliography: "Stress Concentration Factors", by R.E. Peterson

K_{tGROSS}, K_{tNET} & SAF (Stress Adjustment Factor)



$$\sigma_{net} = \sigma_{nom} = \frac{6M}{h^2 t} \qquad Kt_n = \frac{\sigma_p}{\sigma_{net}}$$

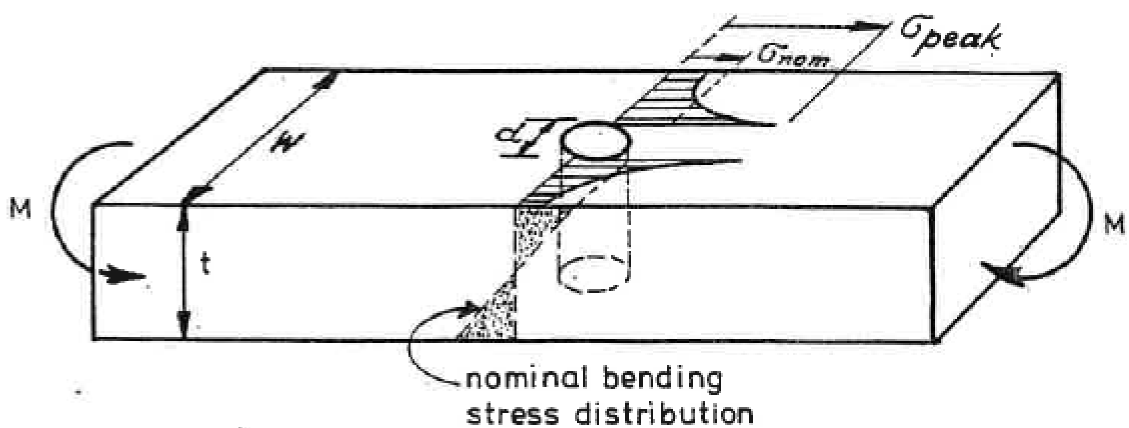
$$\sigma_g = \frac{6M}{W^2 t} \qquad Kt_g = \frac{\sigma_p}{\sigma_g}$$

$$SAF_{net} = \frac{Kt_g}{Kt_{net}} = \left(\frac{W}{h}\right)^2$$

$$\sigma_g = \frac{6M}{Wt^2} \qquad Kt_g = \frac{\sigma_{peak}}{\sigma_g}$$

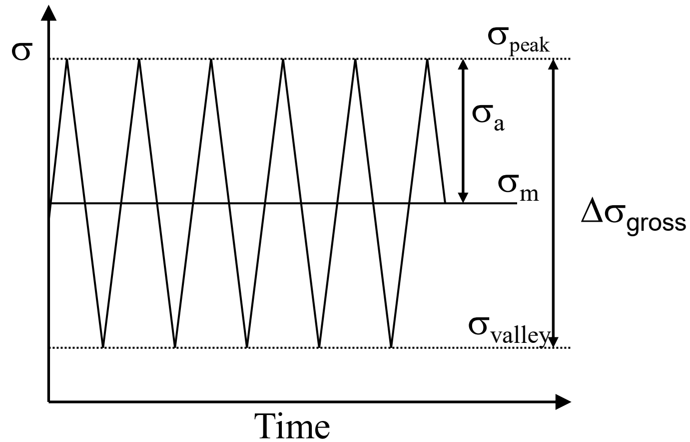
$$\sigma_{net} = \frac{6M}{(W-d)t^2} \qquad Kt_{net} = \frac{\sigma_{peak}}{\sigma_{net}}$$

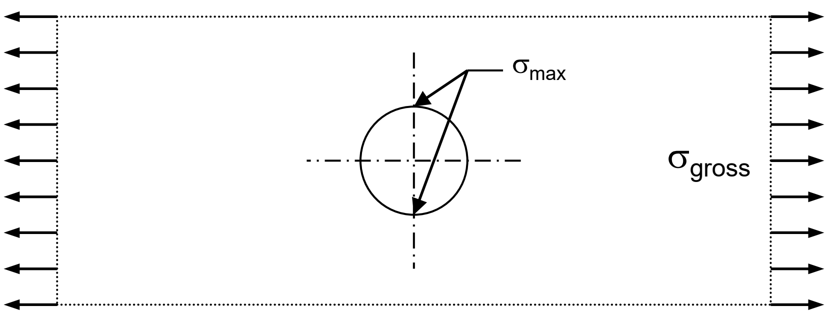
$$SAF_{net} = \frac{Kt_g}{Kt_{net}} = \frac{1}{(1-d/W)}$$



Bibliography: "Stress Concentration Factors", by R.E. Peterson

K_{tGROSS}, K_{tNET} & SAF (Stress Adjustment Factor)





Geometry:
$Kt_{net} = 3$

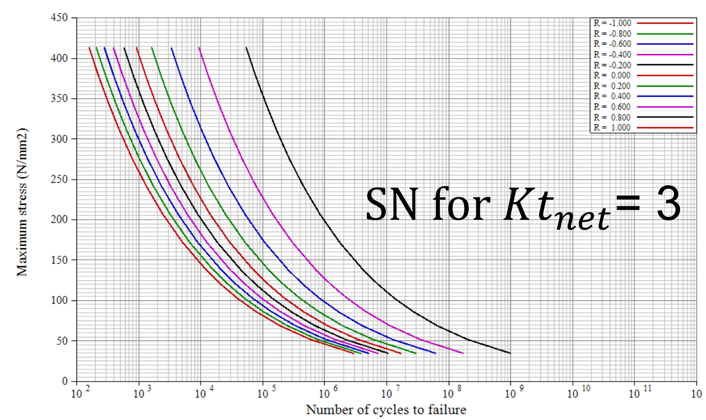
Spectrum in gross stress:
$\Delta\sigma_{gross}$

$\Delta\sigma_{gross}$

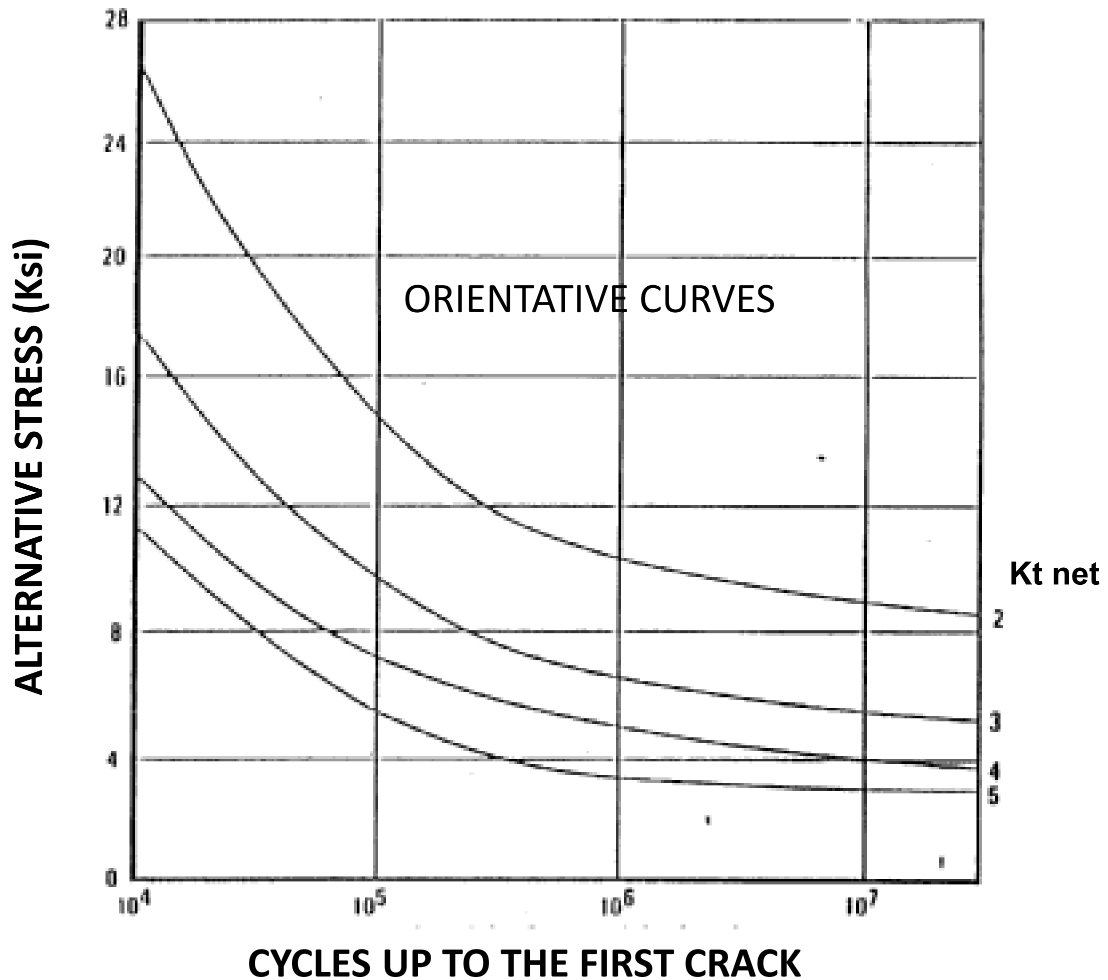
$\Delta\sigma_{max} = Kt_{net} \times \Delta\sigma_{net} = Kt_{net} \times SAF_{net} \times \Delta\sigma_{gross}$

Spectrum in net stress:
$\Delta\sigma_{net} = SAF_{net} \times \Delta\sigma_{gross}$

$\Delta\sigma_{net}$



INFLUENCE OF K_t ON THE SERVICE LIFE UNDER FATIGUE



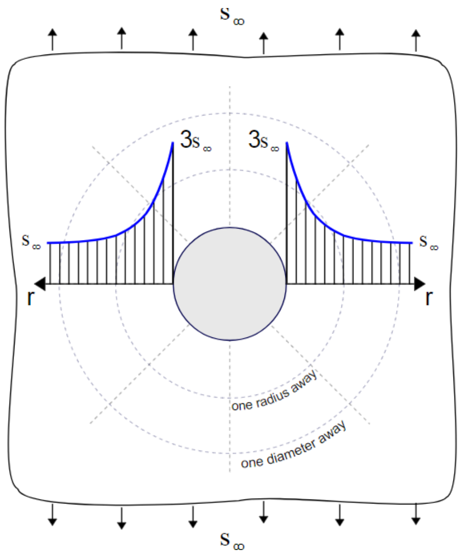
04.2 Stress concentration factors derivation

INTRODUCTION

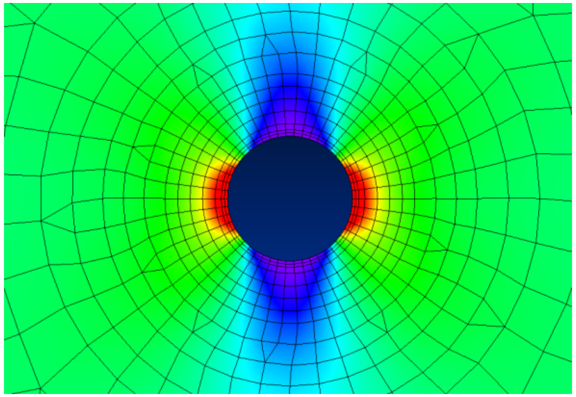
The values for Kt, for a wide variety of geometries and conditions can be obtained from different engineering manuals published. The most commonly used is "STRESS CONCENTRATION FACTORS", by R.E. PETERSON. Other are ESDU or HSB. Values of Kt can be obtained from DFEM as well, especially of complex geometries. Photoelastic methods are very useful to identify areas of stress concentration.

Photoelasticity

Analytical



DFEM



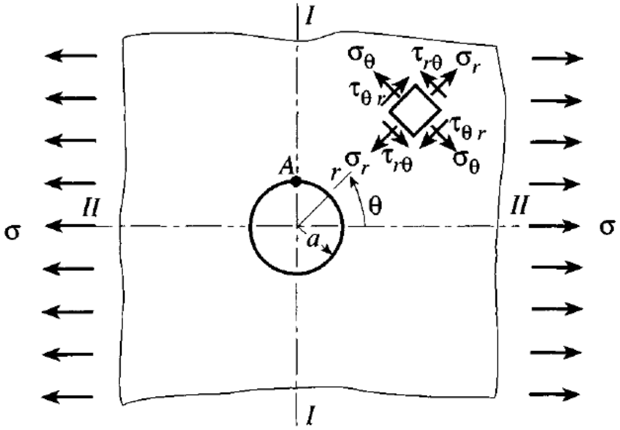
Kt OF HOLES. Open hole in infinite element

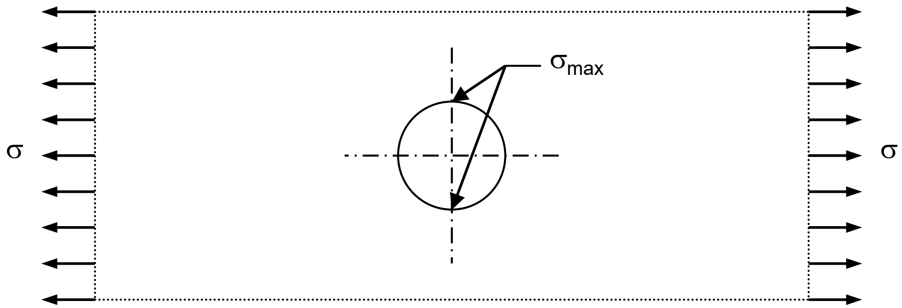
According to Peterson's Book 2nd edition (section 4.3.1), a fundamental case of stress concentration is the study of the stress distribution around a circular hole in an infinite thin element (panel), which is subjected to uniaxial in-plane tension stress (see figure below). In polar coordinates, with applied stress σ the stresses are given as (Timoshenko and Goodier 1970):

$$\sigma_r = \frac{1}{2}\sigma\left(1-\frac{a^2}{r^2}\right)+\frac{1}{2}\sigma\left(1-\frac{4a^2}{r^2}+\frac{3a^4}{r^4}\right)\cos 2\theta$$

$$\sigma_\theta = \frac{1}{2}\sigma\left(1+\frac{a^2}{r^2}\right)-\frac{1}{2}\sigma\left(1+\frac{3a^4}{r^4}\right)\cos 2\theta$$

$$\tau_{r\theta} = -\frac{1}{2}\sigma\left(1+\frac{2a^2}{r^2}-\frac{3a^4}{r^4}\right)\sin 2\theta$$





The maximum stress is found in the hole points perpendicular to the load. The stress concentration factor for the far field is → Kt = 3

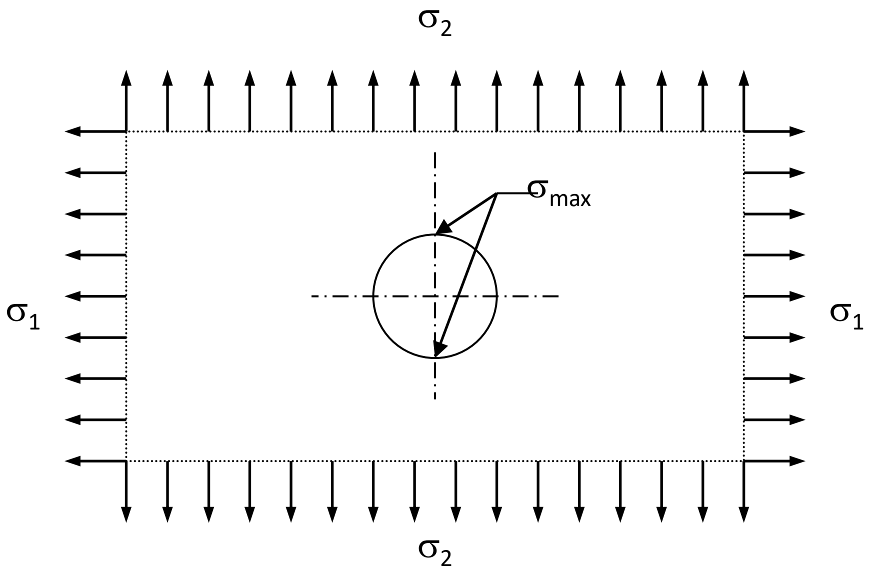
Kt OF HOLES. Open hole in infinite element

In case of biaxial load case, the stress concentration factor for the highest stress is calculated by superposition:

$$\sigma_1 \geq \sigma_2$$

$$Kt = 3 - \frac{\sigma_2}{\sigma_1}$$

$$\sigma_{max} = Kt \cdot \sigma_1$$



Case:

$\sigma_1 \geq \sigma_2$

From previous expressions:

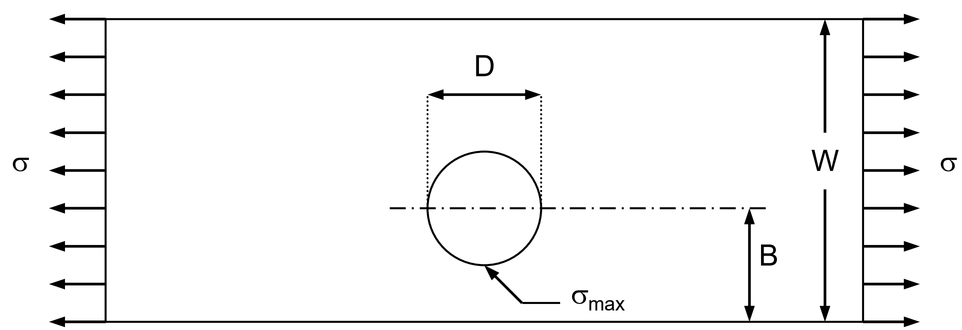
Uniaxial load case (σ_2 = 0) → Kt =3
Biaxial load case ($\sigma_2 = \sigma_1$) → Kt = 2

Kt OF HOLES. Open hole in finite thin element. Axial by-pass stress

The maximum stress is found in the hole points perpendicular to the load closer to the edge. An empirical formula for Ktn to cover the entire D/B range is based on the analytical solutions proposed by Heywood (1952) & Sjöström (1950).

$$Kt_n = 2 + \left(1 - \frac{D}{2 \cdot B}\right)^3$$

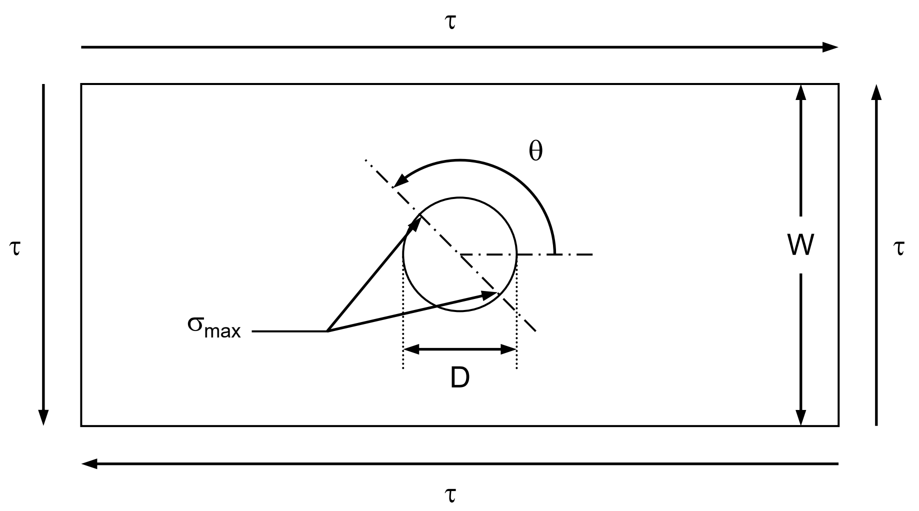
$$\sigma_{net} = \sigma \cdot \frac{\sqrt{1 - \left(\frac{D}{2 \cdot B}\right)^2}}{\left(1 - \frac{D}{2 \cdot B}\right) \cdot \left[1 - \frac{B}{W - B} \cdot \left(1 - \sqrt{1 - \left(\frac{D}{2 \cdot B}\right)^2}\right)\right]}$$



See Peterson's Book 2nd Edition for other configurations:
Charts 4.1 to 4.3.- Kt for the tension of a thin finite or semi-infinite element.
Charts 4.7 to 4.19.- Reinforcement holes in finite and infinite thin element with several states of stresses.

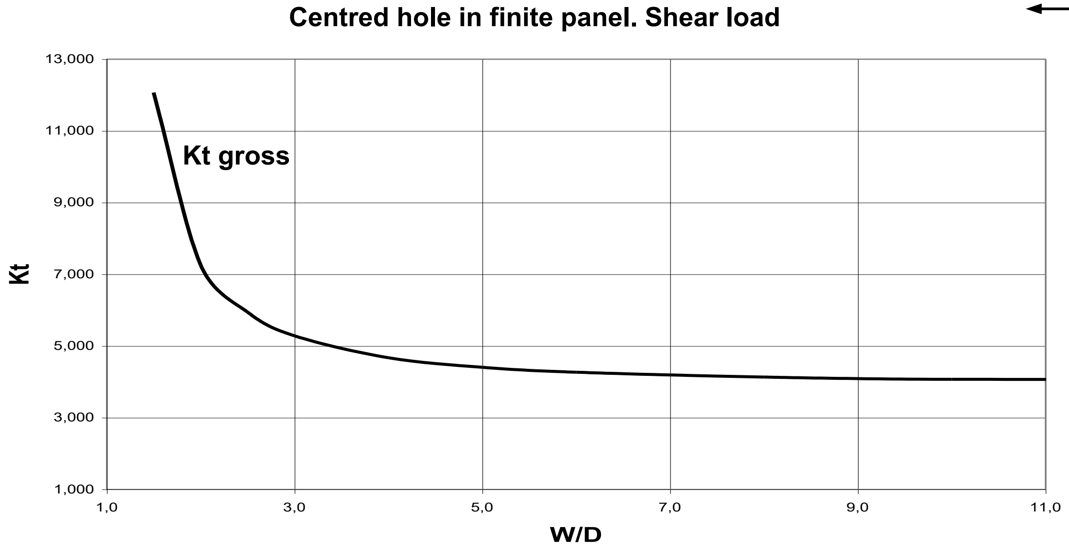
Kt OF HOLES. Open hole in infinite/finite thin element. Shear stress.

The maximum stress is found in the hole points perpendicular to the maximum principal load.



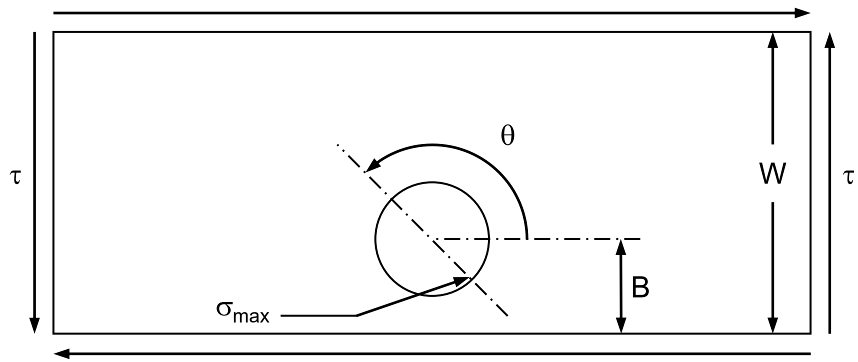
Pure shear load case in infinite panel:
$(\sigma_1 = \tau, \sigma_2 = -\tau) \rightarrow Kt = 4$

Pure shear load case in finite panel:

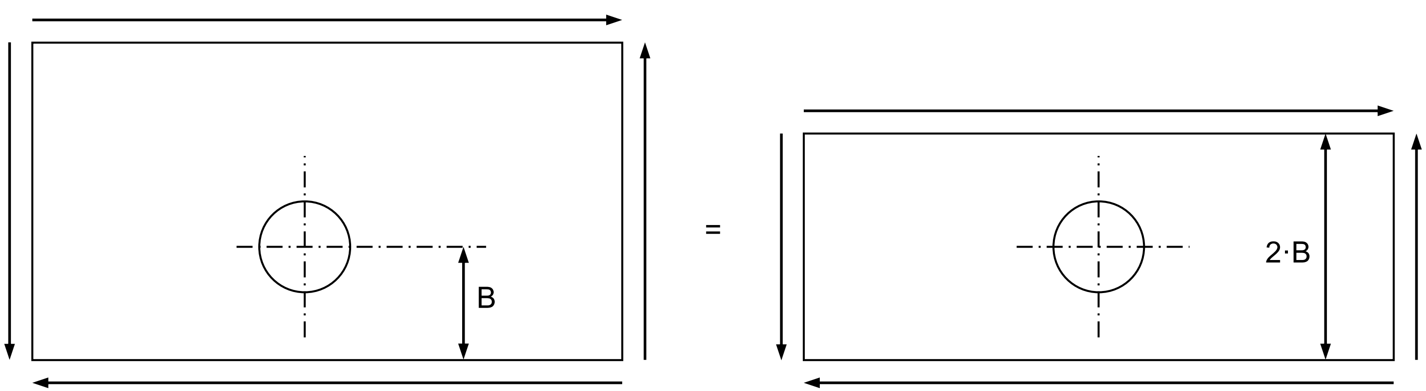


Kt OF HOLES. Open hole in finite thin element. Shear stress.

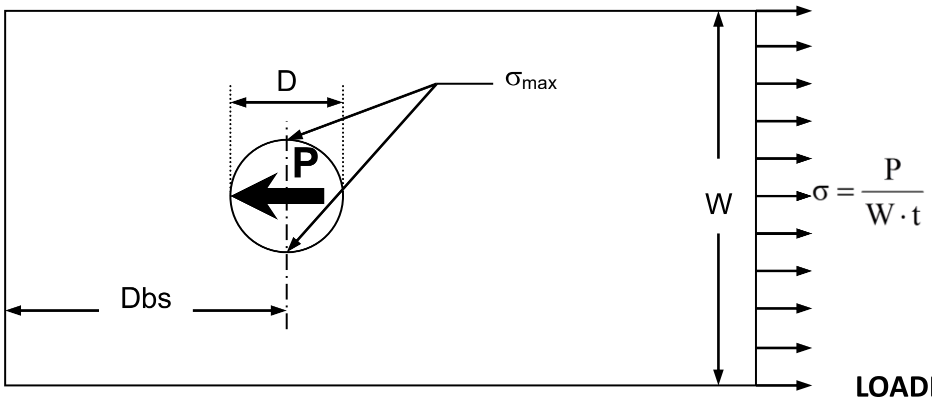
If the hole is not in the center, the maximum stress point is located in the point on the hole closer to the edge and perpendicular to the maximum principal stress.



In order to calculate the stress concentration factor the following hypothesis should be used:



Kt OF HOLES. Loaded hole in finite thin element. Axial load.



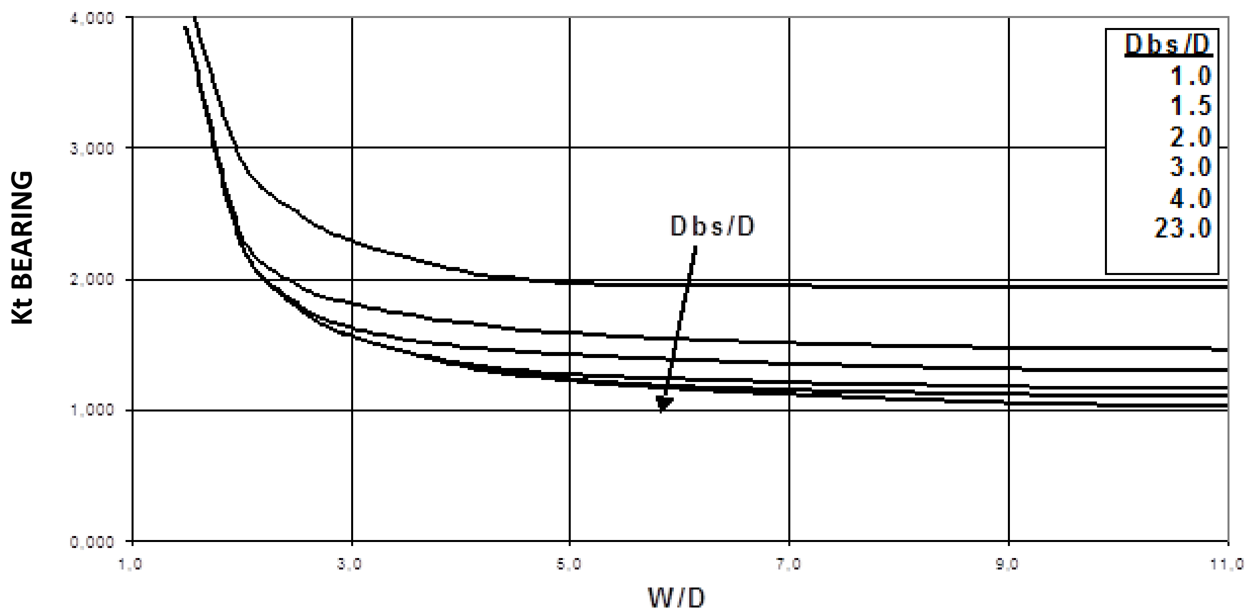
For loaded holes the stress concentration factor is referred to the bearing stress. The maximum stress points are located perpendicularly to the load

$$Kt_{bg} = \frac{\sigma_{max}}{\sigma_{bg}}$$

$$\sigma_{bg} = \frac{P}{D \cdot t}$$

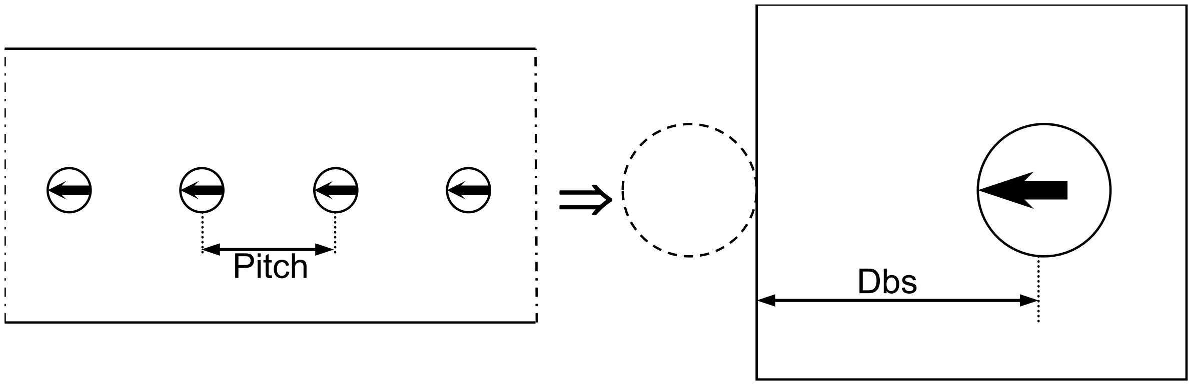
P = Applied load at the hole.
D = Hole diameter.
t = Thickness panel.

LOADED HOLE IN FINITE THIN ELEMENT. AXIAL LOAD.



Kt OF HOLES. Loaded hole in finite thin element. Axial load.

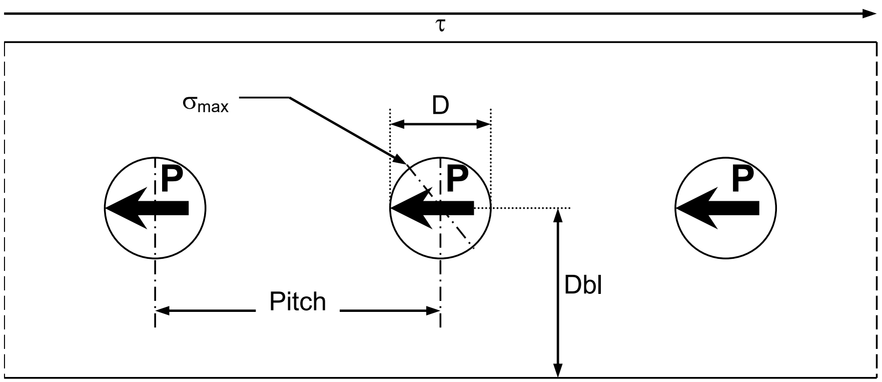
In the case of a row of holes all of them with load transfer, the analysis hypothesis is the following:



$$Dbs = Pitch - \frac{D}{2}$$

$$D = Diameter$$

Kt OF HOLES. Loaded hole in finite thin element. Shear load.

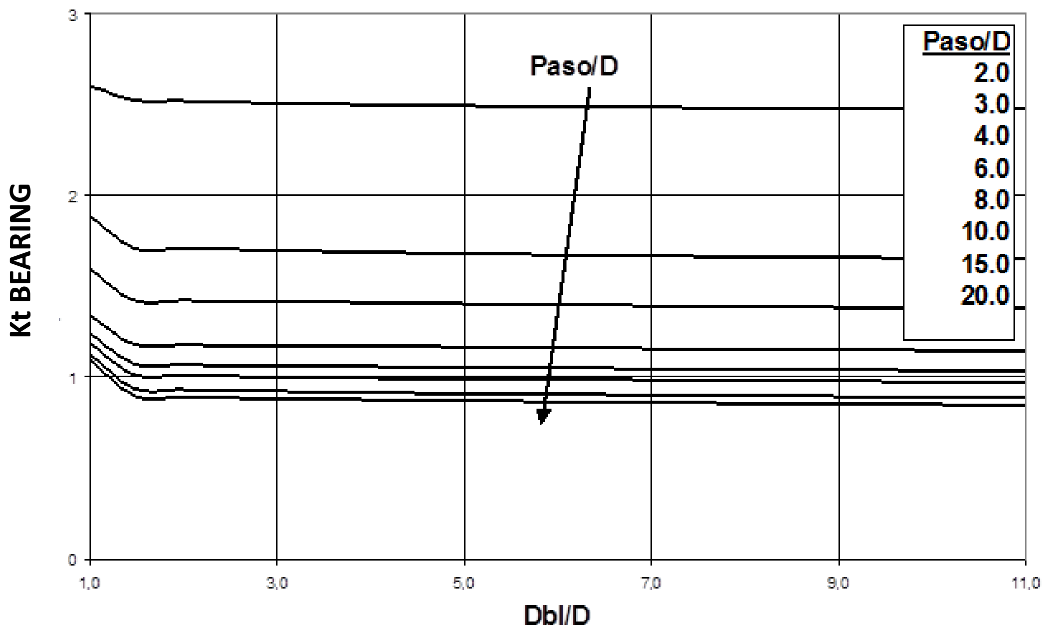


$$\tau = \frac{P}{Pitch \cdot t}$$

$$t = Thickness$$

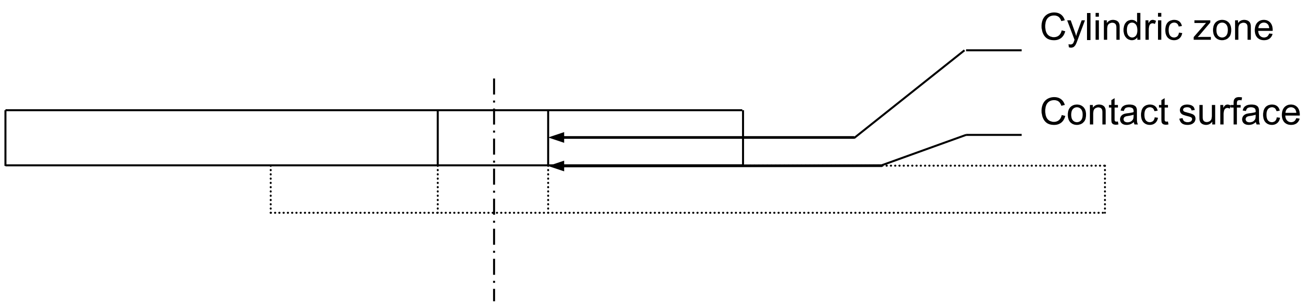
The maximum stress is found in the hole points perpendicular to the maximum principal stress and closer to the reaction edge of the hole.

LOADED HOLE IN FINITE THIN ELEMENT. SHEAR LOAD.

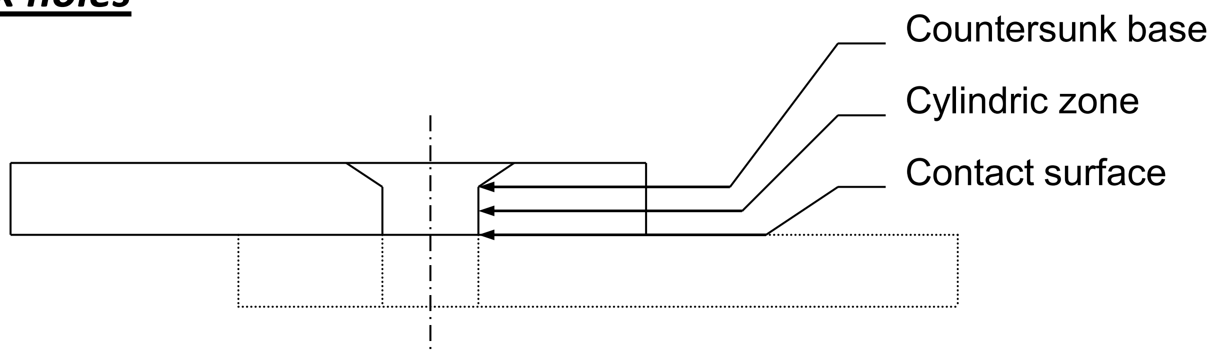


ADDITIONAL EFFECTS OVER Kt.

Cilindrical holes

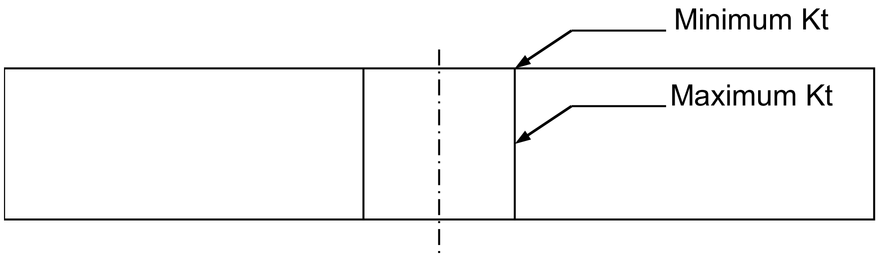


Countersunk holes

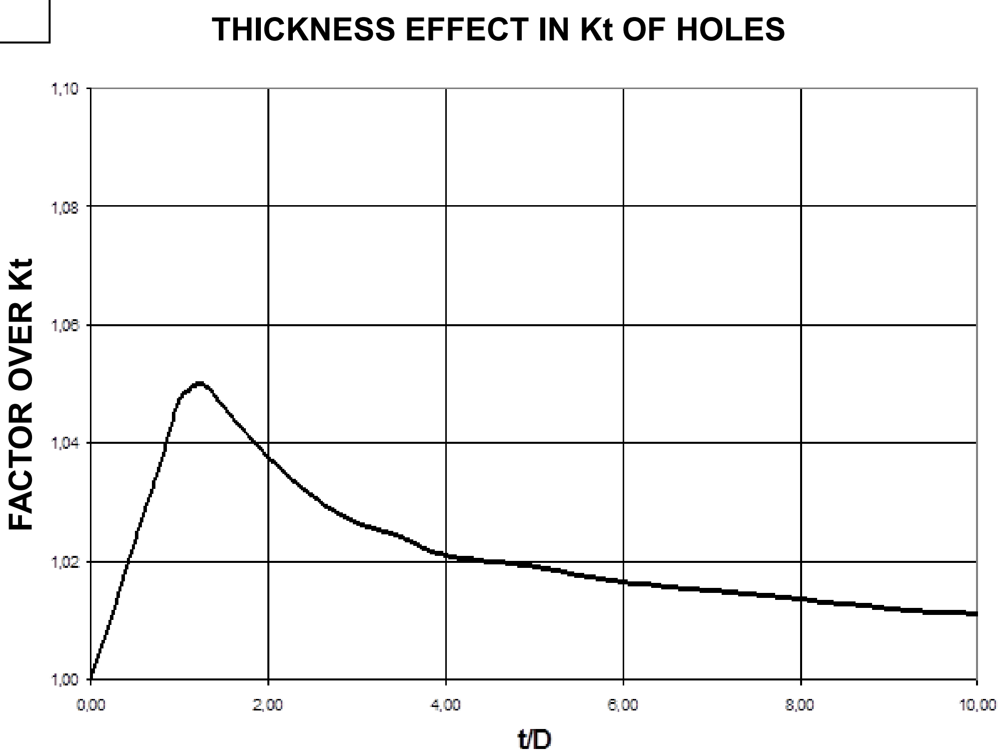


ADDITIONAL EFFECTS OVER Kt.

Thickness effect



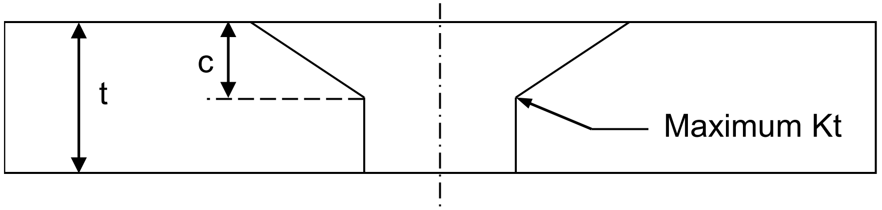
The maximum Kt is located in the middle plane, but this effect should not be combined with the fastener bending effect, which produces maximum Kt at the surface.



ADDITIONAL EFFECTS OVER Kt.

Countersunk effect

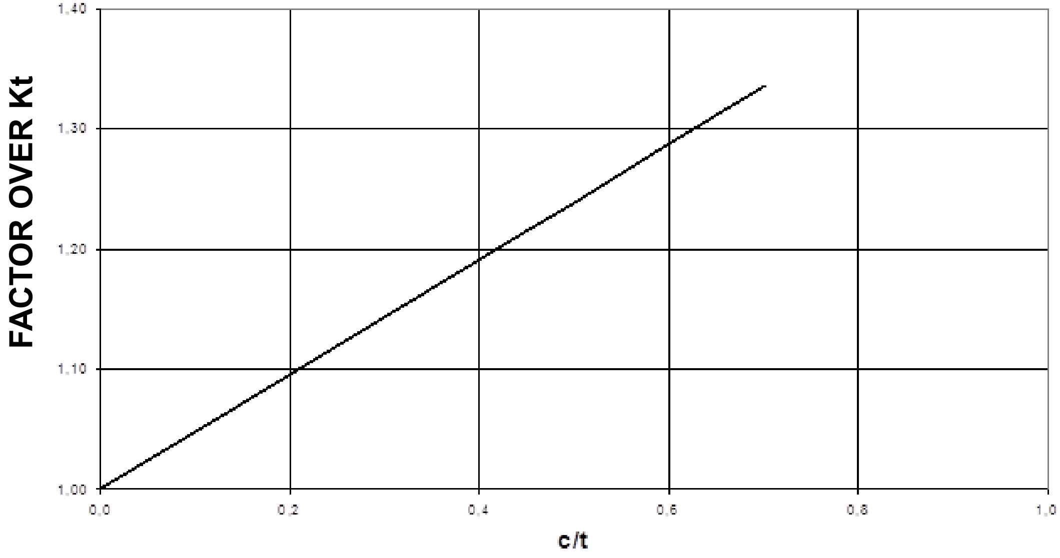
The maximum stress is located in the intersection between countersunk zone and cylindrical zone.



Three regions are defined:

- Design region: c/t≤0.7
- Repair region: 0.7<c/t≤0.9 (min cylindric zone thickness is 0.25mm)
- Not allowed region: 0.9<c/t<1

COUNTERSUNK EFFECT IN Kt OF HOLES



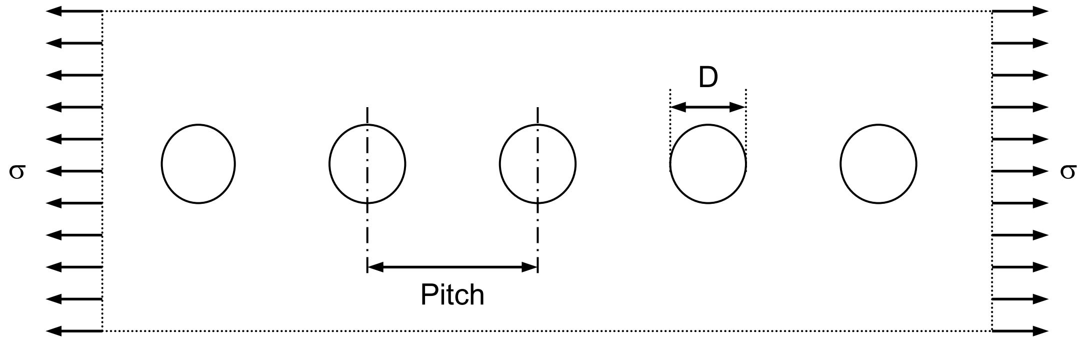
Different authors calculate the countersunk effect depending on c/t and D/t ratios, based on the study of different solutions (Northrop, Shivakumar, etc)

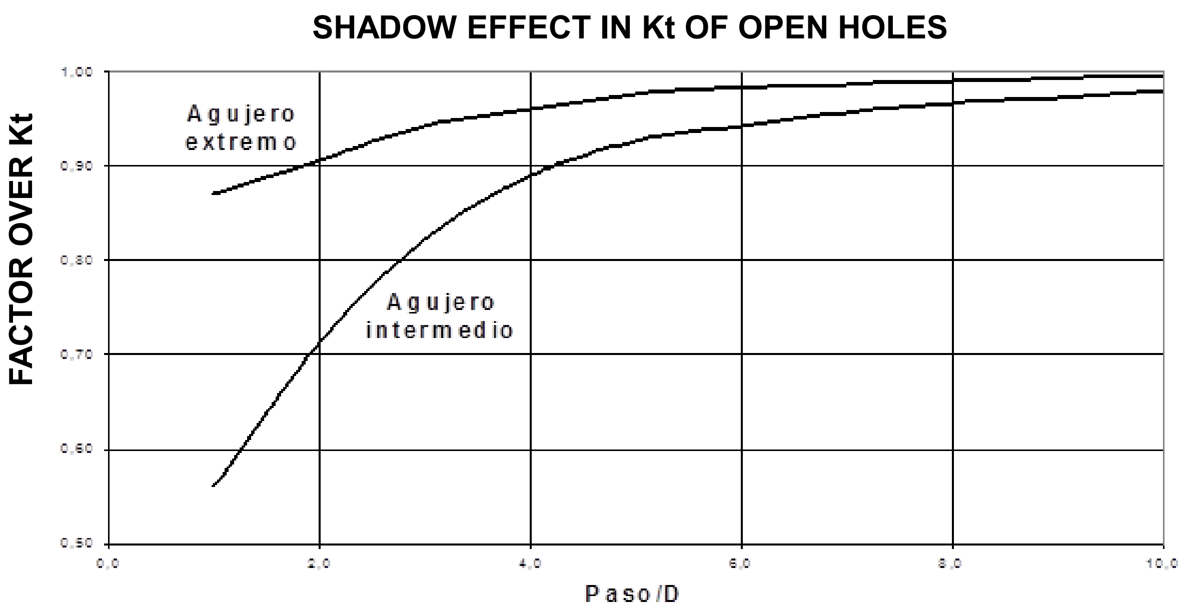
ADDITIONAL EFFECTS OVER Kt.

Shadow effect

In case of the row of rivets the interaction between holes produces a reduction of the Kt due to called shadow effect. This effect is only applied when the load is applied in the row direction. As in real situations the load is usually a combination of loads in different directions this effect is not considered conservatively.

A solution for shadow effect can be obtained from Peterson.



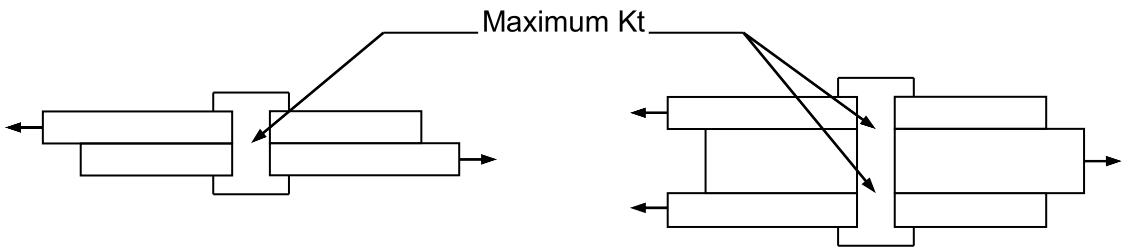


ADDITIONAL EFFECTS OVER Kt.

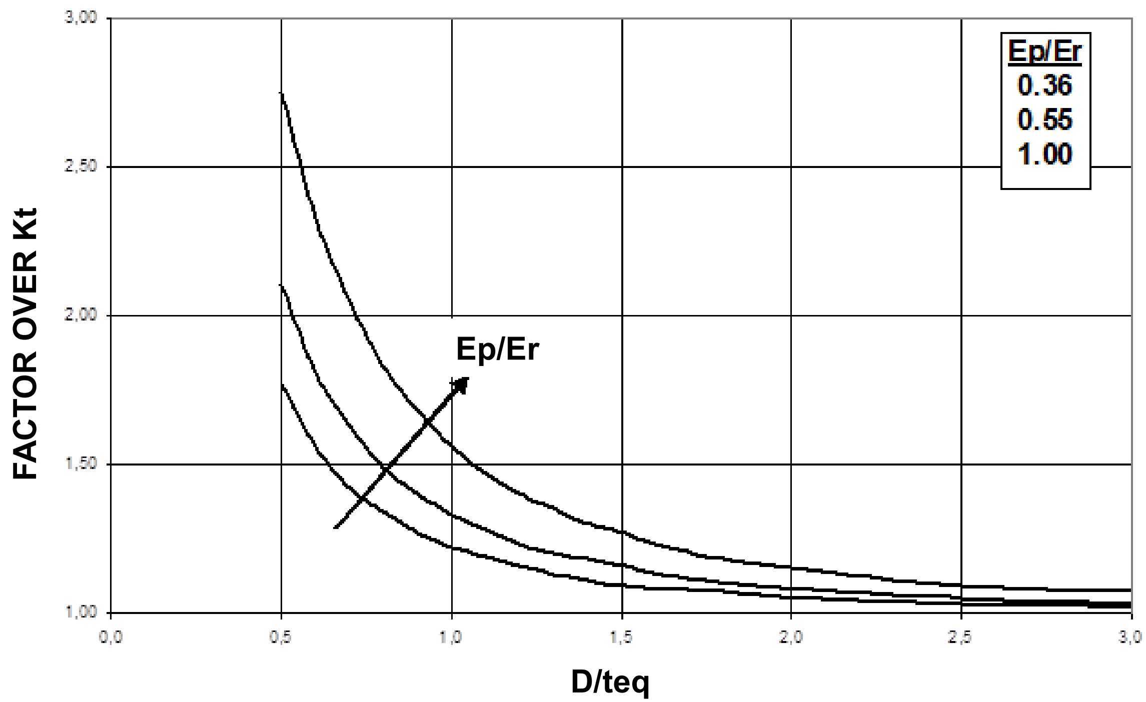
Pin bending effect

The maximum stress is located at contact surface.

This effect must not be combined with thickness effect and countersunk effect which produces maximum Kt at the middle plane).



PIN BENDING EFFECT IN Kt OF OPEN HOLES



ADDITIONAL EFFECTS OVER Kt.

Pin bending effect

The fastener bending effect is tabulated with the following parameters:

D = diameter hole

Ep = Panel Young modulus

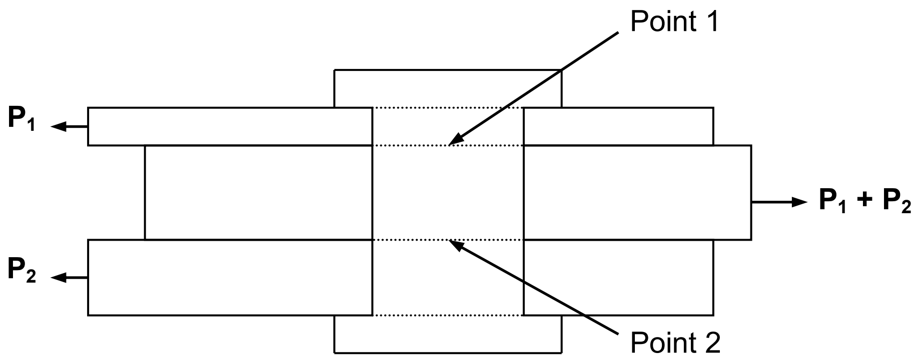
Er = Fastener Young modulus

t_{eq} = equivalent thickness

- Joint to simple shear: t_{eq} = panel nominal thickness.
- Joint to double shear
  - Extreme elements: teq = Real thicknerss.
  - Central element:

$\rightarrow$ Point 1: $t_{eq} = t \cdot \dfrac{P_1}{P_1 + P_2}$

$\rightarrow$ Point 2: $t_{eq} = t \cdot \dfrac{P_2}{P_1 + P_2}$

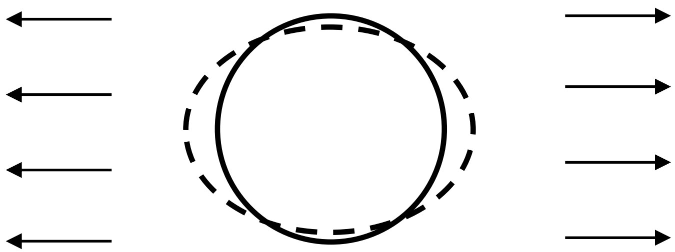


ADDITIONAL EFFECTS OVER Kt.

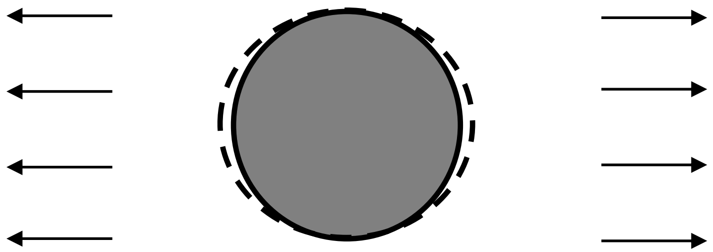
<u>Filling effect</u>

The filling is a beneficial effect from fatigue point view because the stress concentration factor is reduced as a consequence of preventing the free hole deformation.

In a conservative way, <u>the filling effect must not be used in any analysis</u> except for correlating test or real cracks evidences.

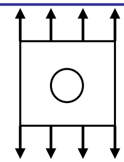
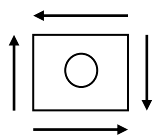
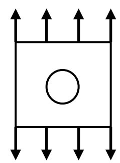
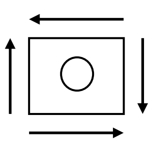


UNFILLED HOLED

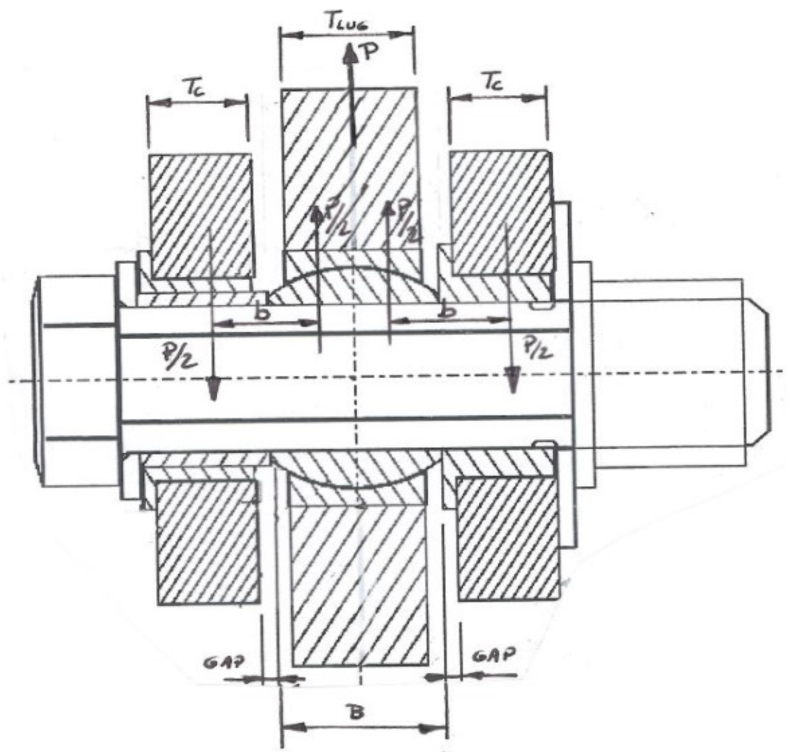


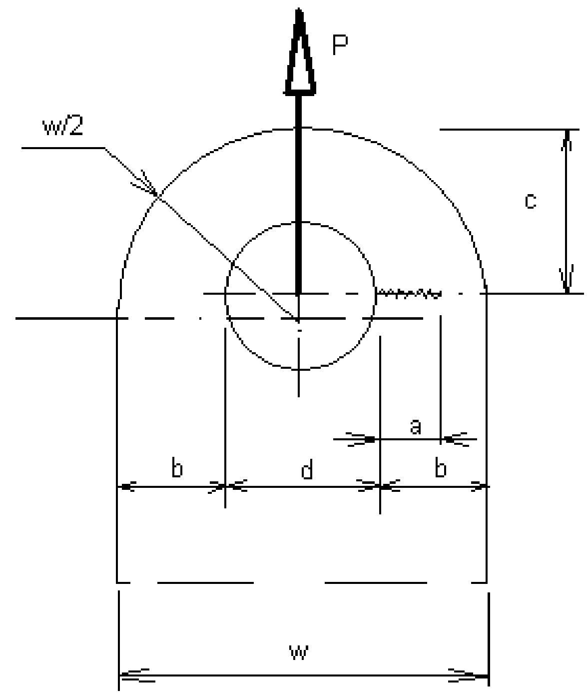
FILLED HOLED

ADDITIONAL EFFECTS OVER Kt.

	Load		Surface	By-Pass load case (By-Pass Kt)				Load transfer case (Bearing Kt)			
				Shadow	**Thick.**	**Counter**	**Pin bend.**	**Shadow**	**Thick.**	**Counter**	**Pin bend.**
Cylindric hole	**Axial**		Cylindric Zone	**Yes**	**Yes**	No	No	No	**Yes**	No	No
			Contact Surface	**Yes**	No	No	No	No	No	No	**Yes**
	Shear		Cylindric Zone	**Yes**	**Yes**	No	No	No	**Yes**	No	No
			Contact Surface	**Yes**	No	No	No	No	No	No	**Yes**
Countersunk hole	**Axial**		Cylindric Zone	**Yes**	**Yes**	No	No	No	**Yes**	No	No
			Counter. Base	**Yes**	No	**Yes**	No	No	No	**Yes**	No
			Contact Surface	**Yes**	No	No	No	No	No	No	**Yes**
	Shear		Cylindric Zone	**Yes**	**Yes**	No	No	No	**Yes**	No	No
			Counter. Base	**Yes**	No	**Yes**	No	No	No	**Yes**	No
			Contact Surface	**Yes**	No	No	No	No	No	No	**Yes**

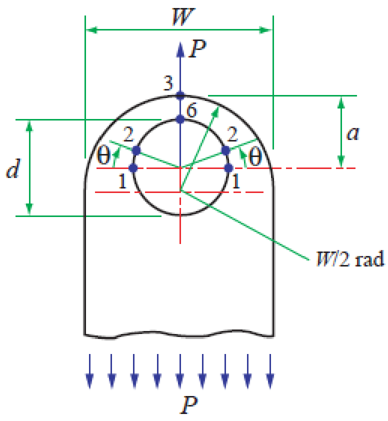
Kt OF LUGS

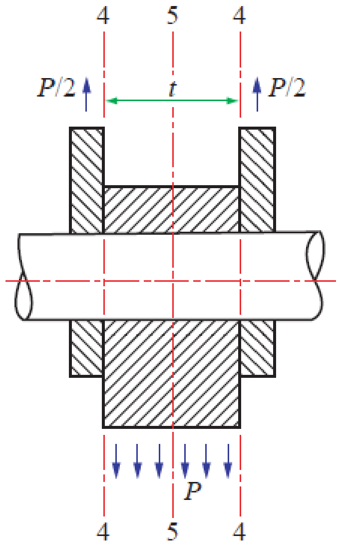




Kt OF LUGS

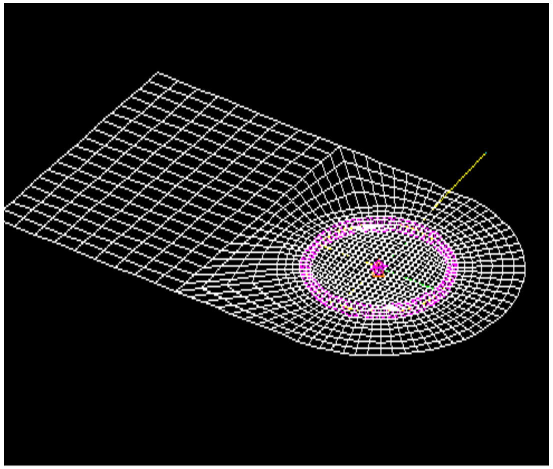
Kt of lugs can be calculated basically in two different ways:





- From engineering manuals published (e.g. ESDU 81006, 81029). Recommended for simple straight lugs axially loaded.

- By FEM analysis: recommended for fittings with complex geometries and lugs non axially loaded. A detailed meshing must be used near the inner diameter to obtain the maximum stress correctly. 1D elements in 2D models and 2D elements in 3D models must be used to obtain the maximum stress in the edge of the inner diameter.



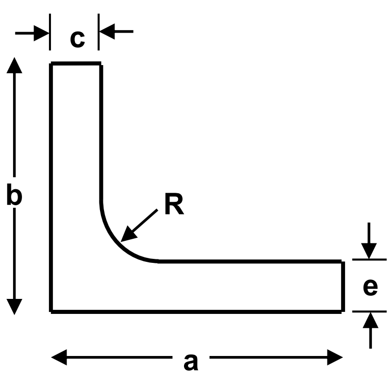
- Additional correction factors might be used in both cases (Pin bending effect, angle effect, clearance fit effect, lug thickness effect, ...)

Kt OF FILLET RADIUS

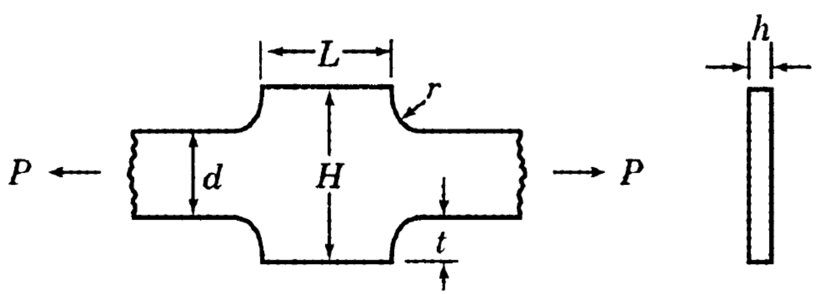
<u>A) Tension at fillet radius</u>

This calculation is based on the Peterson's Book chart 3.2. Kt fillet radius is calculated assuming a symmetric configuration.

<u>Analysis to be performed</u>



<u>Symmetric problem (from Peterson's)</u>



The geometrical relationships between both configurations are:

<u>Geometrical Relationships:</u>

$r = R$; $d = 2 \cdot e$; $L = 2 \cdot c$; $H = 2 \cdot b$; $t = b - e$

Kt OF FILLET RADIUS

From previous relationships the Kt is calculated according to the following mathematics expressions:

$$K_t = C_1 + C_2\left(\frac{2t}{H}\right) + C_3\left(\frac{2t}{H}\right)^2 + C_4\left(\frac{2t}{H}\right)^3$$

$$\text{where } \frac{L}{H} > -1.89\left(\frac{r}{d} - 0.15\right) + 5.5$$

	$0.1 \le t/r \le 2.0$	$2.0 \le t/r \le 20.0$
C_1	$1.006 + 1.008\sqrt{t/r} - 0.044t/r$	$1.020 + 1.009\sqrt{t/r} - 0.048t/r$
C_2	$-0.115 - 0.584\sqrt{t/r} + 0.315t/r$	$-0.065 - 0.165\sqrt{t/r} - 0.007t/r$
C_3	$0.245 - 1.006\sqrt{t/r} - 0.257t/r$	$-3.459 + 1.266\sqrt{t/r} - 0.016t/r$
C_4	$-0.135 + 0.582\sqrt{t/r} - 0.017t/r$	$3.505 - 2.109\sqrt{t/r} + 0.069t/r$

Kt graphics for the stress concentration factor at fillet radius under tension load cases can be seen in Peterson’s Book charts 3.2a, 3.2b, 3.2c and 3.2d.

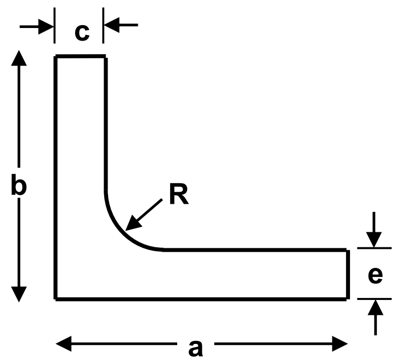
STRESS CONCENTRATION FACTORS

Kt OF FILLET RADIUS

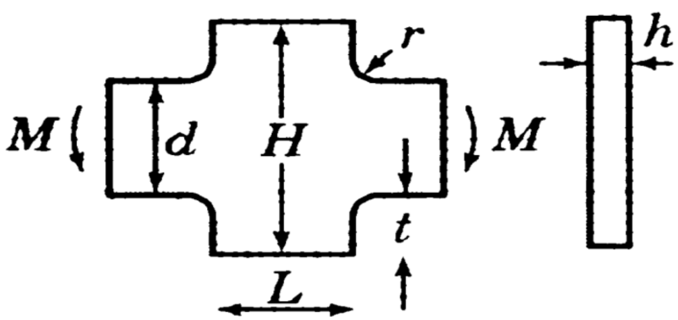
A) Bending at fillet radius

This calculation is based on the Peterson's Book chart 3.8. Kt fillet radius is calculated assuming a symmetric configuration.

Analysis to be perform



Symmetric problem (from Peterson's)



The geometrical relationships between both configurations are:

Geometrical Relationships:

r = R; d = 2 · e; L = 2 · c; H = 2 · b; t = b – e

Kt OF FILLET RADIUS

From previous relationships the Kt is calculated according to the following mathematics expressions:

$$K_t = C_1 + C_2\left(\frac{2t}{H}\right) + C_3\left(\frac{2t}{H}\right)^2 + C_4\left(\frac{2t}{H}\right)^3$$

$$\text{where } \frac{L}{H} > -2.05\left(\frac{r}{d} - 0.025\right) + 2.0$$

	$0.1 \le t/r \le 2.0$	$2.0 \le t/r \le 20.0$
C_1	$1.006 + 0.967\sqrt{t/r} + 0.013t/r$	$1.058 + 1.002\sqrt{t/r} - 0.038t/r$
C_2	$-0.270 - 2.372\sqrt{t/r} + 0.708t/r$	$-3.652 + 1.639\sqrt{t/r} - 0.436t/r$
C_3	$0.662 + 1.157\sqrt{t/r} - 0.908t/r$	$6.170 - 5.687\sqrt{t/r} + 1.175t/r$
C_4	$-0.405 + 0.249\sqrt{t/r} - 0.200t/r$	$-2.558 + 3.046\sqrt{t/r} - 0.701t/r$

Kt graphics for the stress concentration factor at fillet radius under bending load cases can be seen in Peterson’s Book charts 3.8a, 3.8b and 3.2c.

PRINCIPLE OF SUPERPOSITION

The stress concentration factors are dependent on the geometry of the problem and the loading state. There are studies and publications from which Kt factors can be obtained for different simple configurations (e.g. Peterson). In a more complex case, the different simple solutions can be superimposed for each of the geometric effects and real load application.

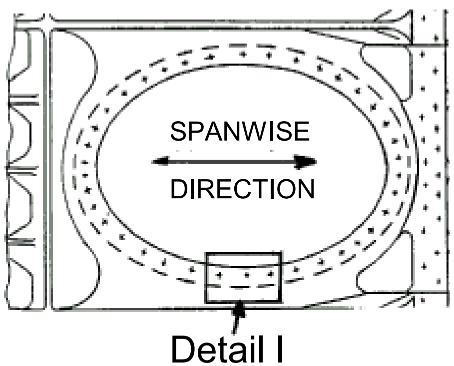
Superposition of geometric effects. The final stress concentration factor will be the product of the different factors for each of the simple effects:

$$Kt = Kt_1 \cdot Kt_2 \cdot \cdots \cdot Kt_n$$

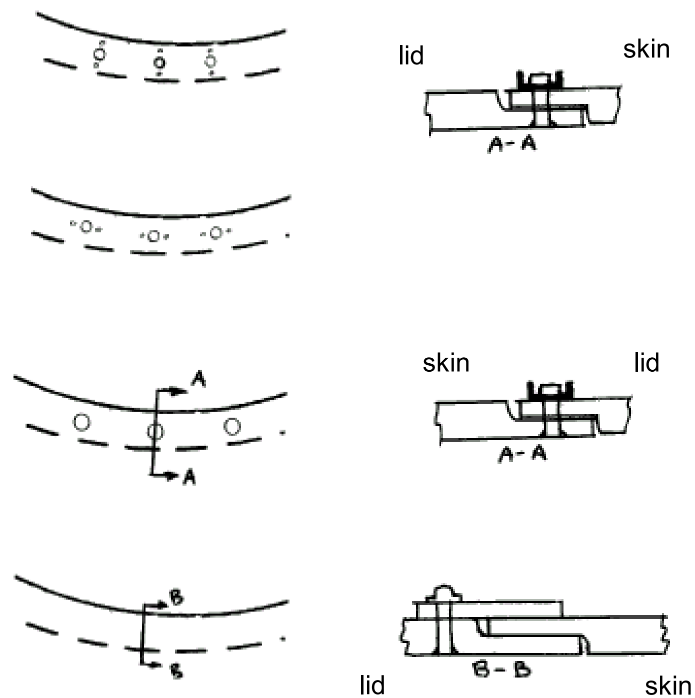
Superposition of load cases. The final factor will be the sum of the different factors for each of the load cases:

$$Kt = Kt_1 + Kt_2 + \cdots + Kt_n$$

SUPERPOSITION OF GEOMETRIC EFFECTS



MAN-HOLE



DETAIL I

$K_{TOTAL} = K1 \times K2 \times K3 \times \ldots$

WORST

Kt1 = K_t (Manhole)
&
Kt2 = K_t (Bolt)
&
Kt3 = K_t (Nut Rivet)

BAD

Kt1 = K_t (Manhole)
&
Kt2 = K_t (Bolt)

BETTER

Kt1 = K_t (Manhole)
&
Kt2 = K_t (Bolt)

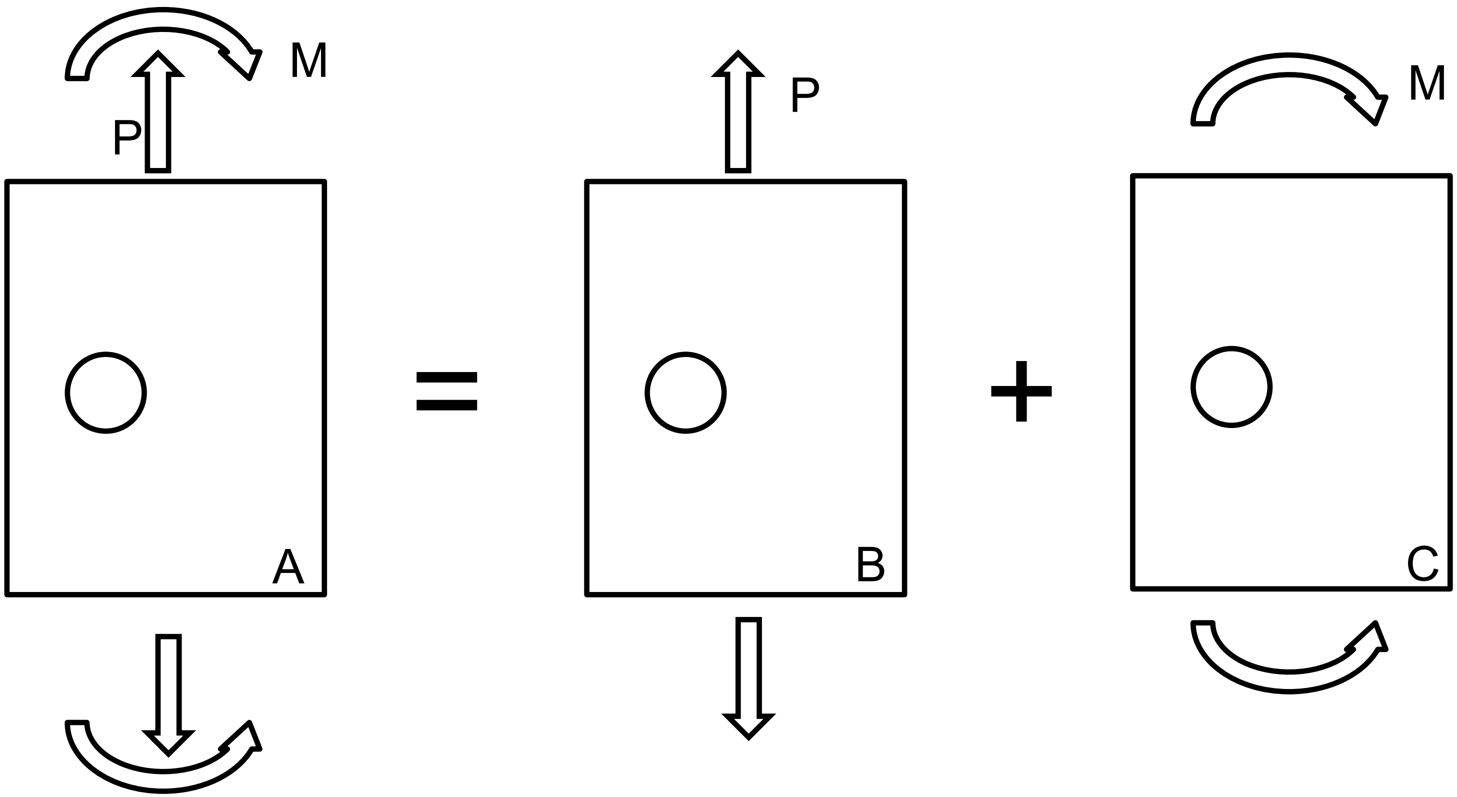
BEST

Kt1 = K_t (Manhole)

SUPERPOSITION OF LOAD CASES

Axial + bending problem

A flat plate with a hole is subjected to a combination of axial and bending loads:

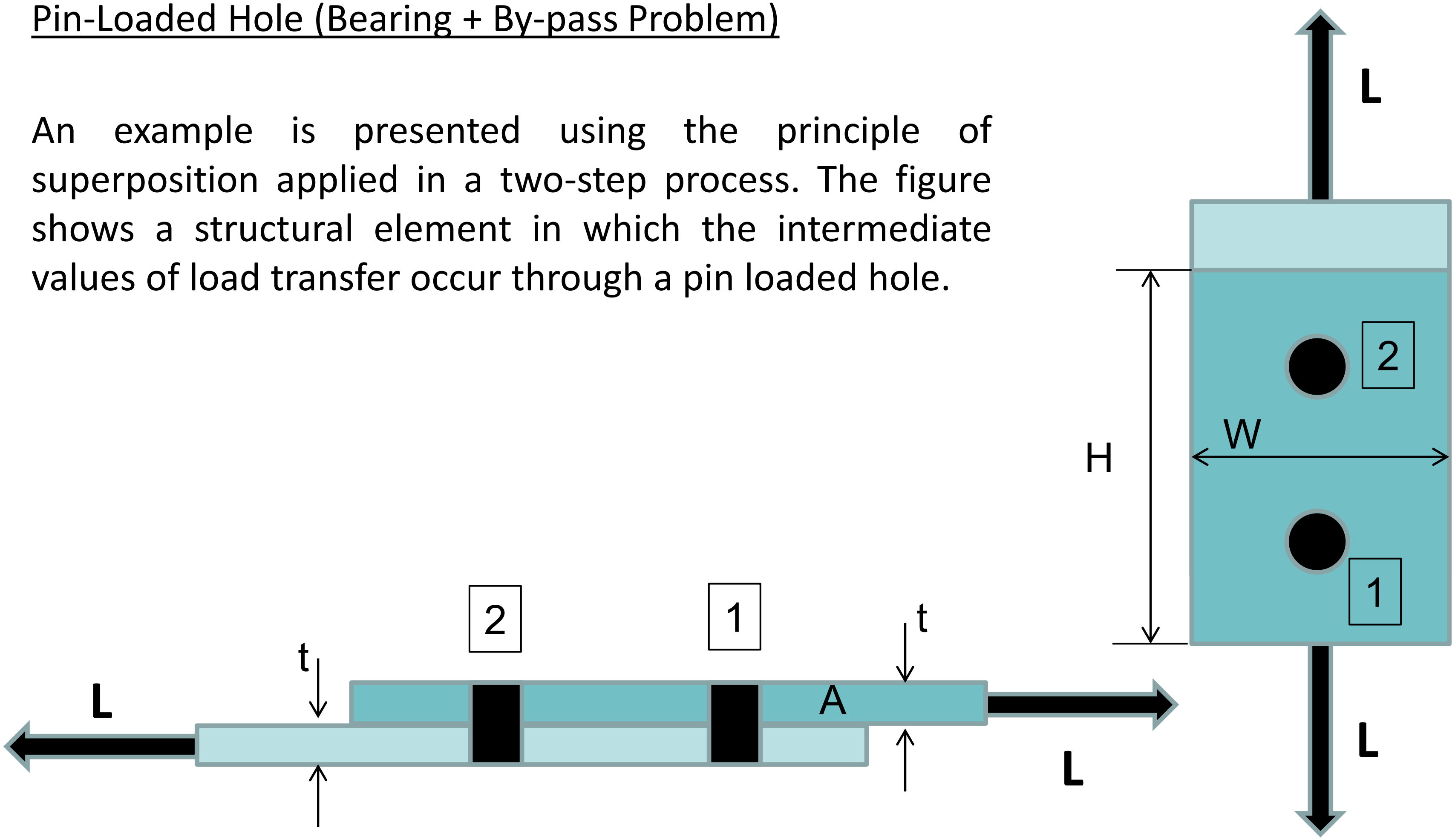


$$K_{tA} = K_{tB} + K_{tC}$$

SUPERPOSITION OF LOAD CASES

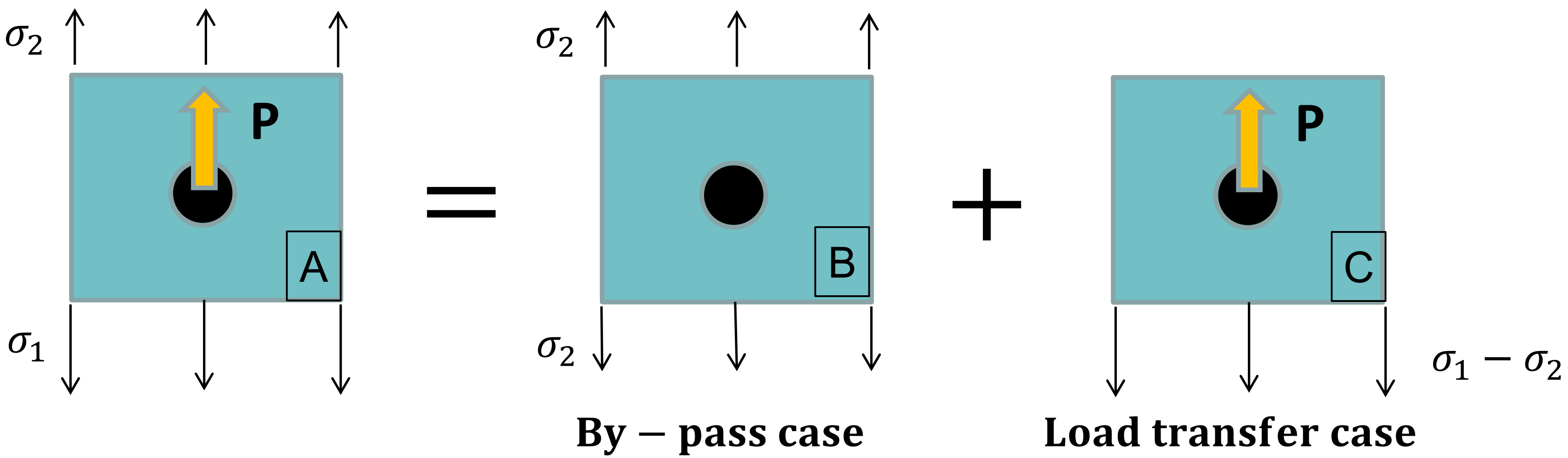
Pin-Loaded Hole (Bearing + By-pass Problem)

An example is presented using the principle of superposition applied in a two-step process. The figure shows a structural element in which the intermediate values of load transfer occur through a pin loaded hole.



SUPERPOSITION OF LOAD CASES

The stress concentration factor of hole 1 of specimen A can be calculated as superposition of two load cases, by-pass (B) and load transfer (C). The Kt's only can be summed when they are refered to the same stress.

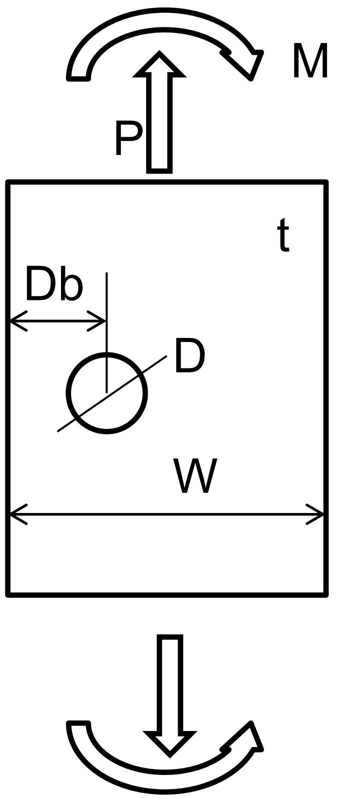


$$K_{tA} = K_{tB} + K_{tC}$$

04.3 Worked examples

EXERCISE 1

Calculate the maximum stress and locate the point of maximum stress. ¿How would you reduce the maximum stress?



W= 40 mm

D = 4 mm

Db= 10 mm

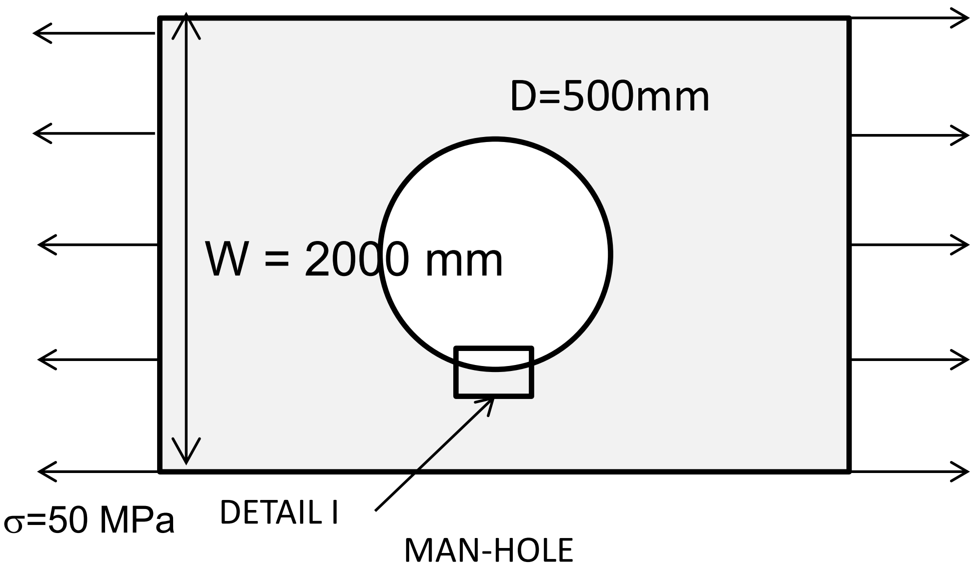
t = 2 mm

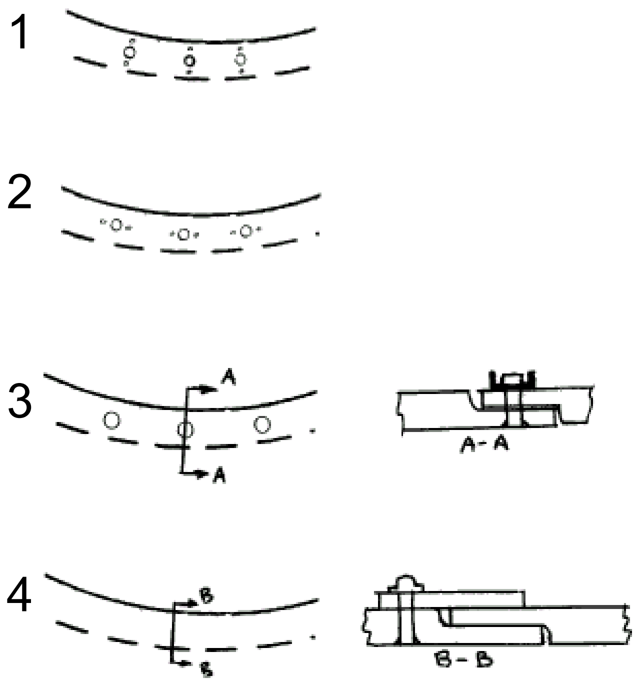
P = 2000 N

M = 20000 Nmm

EXERCISE 2.

Calculate location of the maximum stress, the Kt and maximum stress for the 4 configurations. The bolt holes are 5 mm diameter and the nut attachment holes are 2.5 mm diameter. The bolt holes centers are located at 20 mm from the manhole edge and the nut holes centers are located at 5 mm from the bolt hole edge.

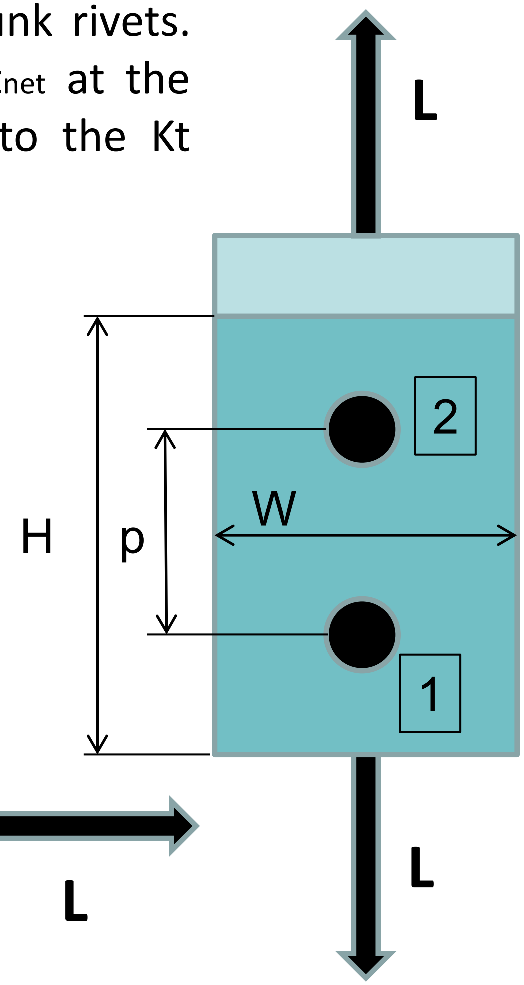




DETAIL I

EXERCISE 3

The joint in the figure is made by means of 2 countersunk rivets. Identify the critical location for fatigue. Calculate the Kt_{net} at the critical location. Apply the following additional factors to the Kt appropriately.



t= 2 mm

W = 20 mm

H= 50 mm

D = 4 mm

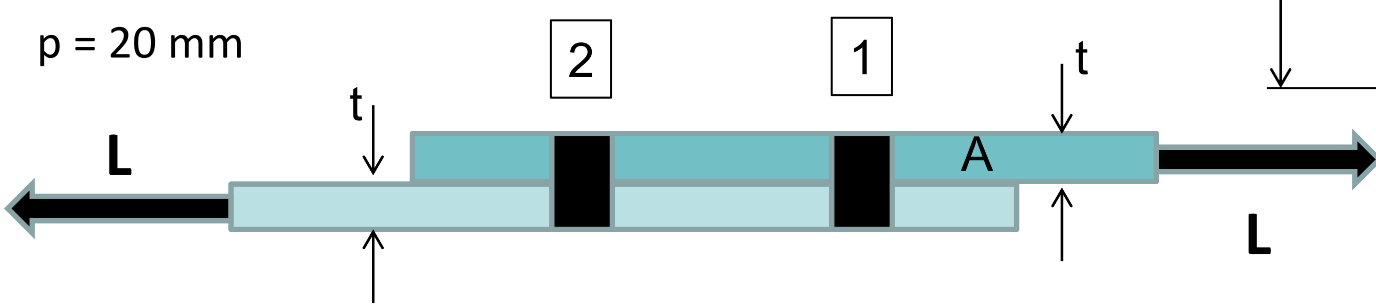
L= 5000 N

p = 20 mm

Kt_{pb} = 1.15 (pin bending)

Kt_{th} = 1.0235 (thickness)

Kt_{csk}=1.2297 (countersunk)



EXERCISE 4

CALCULATE THE Kt_{net} AND Kt_{gross} OF THE FOLLOWING LUG. USE ESDU81006

Location: Male lug at typical wing to fuselage attachment.

There are **two lugs** which are identical, each one with half load. If one lug fails the other one can carry the total load.

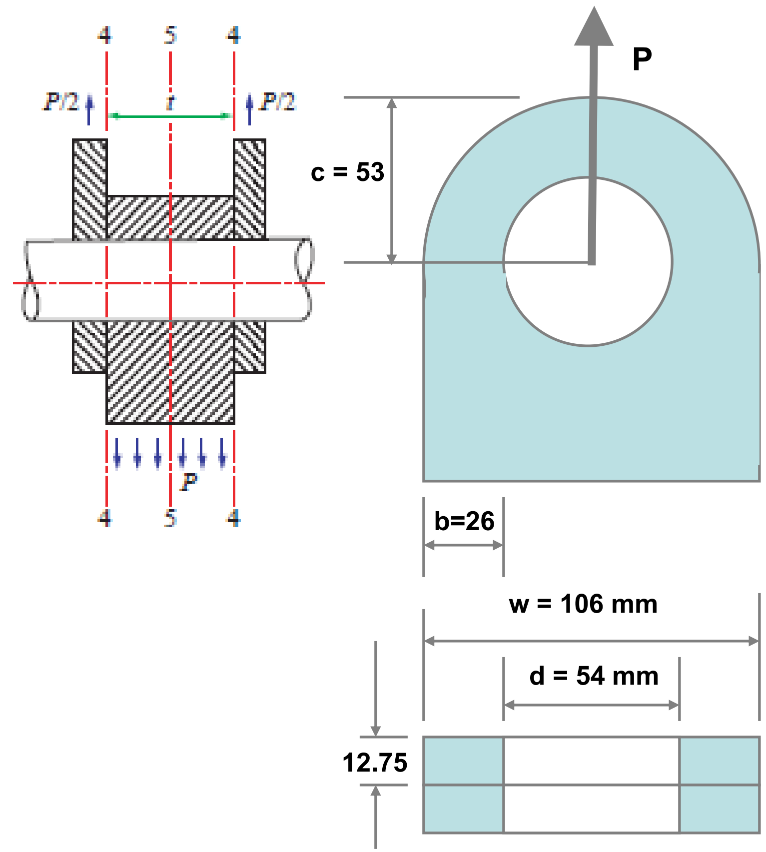
Each lug thickness is **12.75 mm** and the material **7075-T7351.** The pin material is **steel.**

Lug hole diameter is: $\phi_{lug} = \mathbf{54^{+0.10}_{+0.01}}$ **mm**

Pin diameter is: $\phi_{pin} = \mathbf{54^{-0.08}_{-0.01}}$ **mm**

$\sigma_{gross} = P/(W \cdot 2t)$
$SAF = \sigma_{gross}/P = 1/(W \cdot 2t) = 3.7 \cdot 10^{-4}$

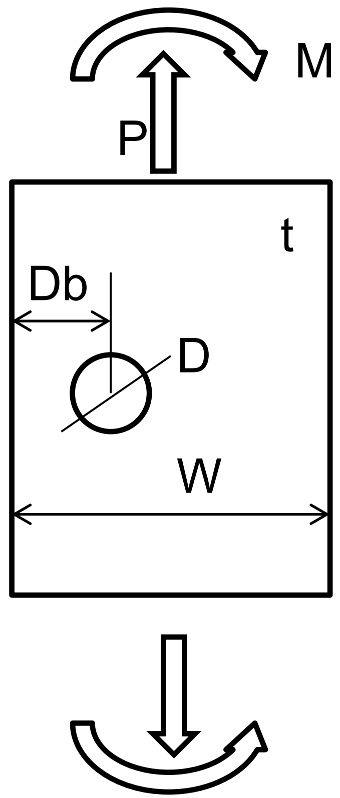
$$e = \frac{(0.10 + 0.08) \cdot 100}{54} = 0.333$$



04.4 Worked examples (solutions)

EXERCISE 1

Calculate the maximum stress and locate the point of maximum stress. ¿How would you reduce the maximum stress?



W= 40 mm

D = 4 mm

Db= 10 mm

t = 2 mm

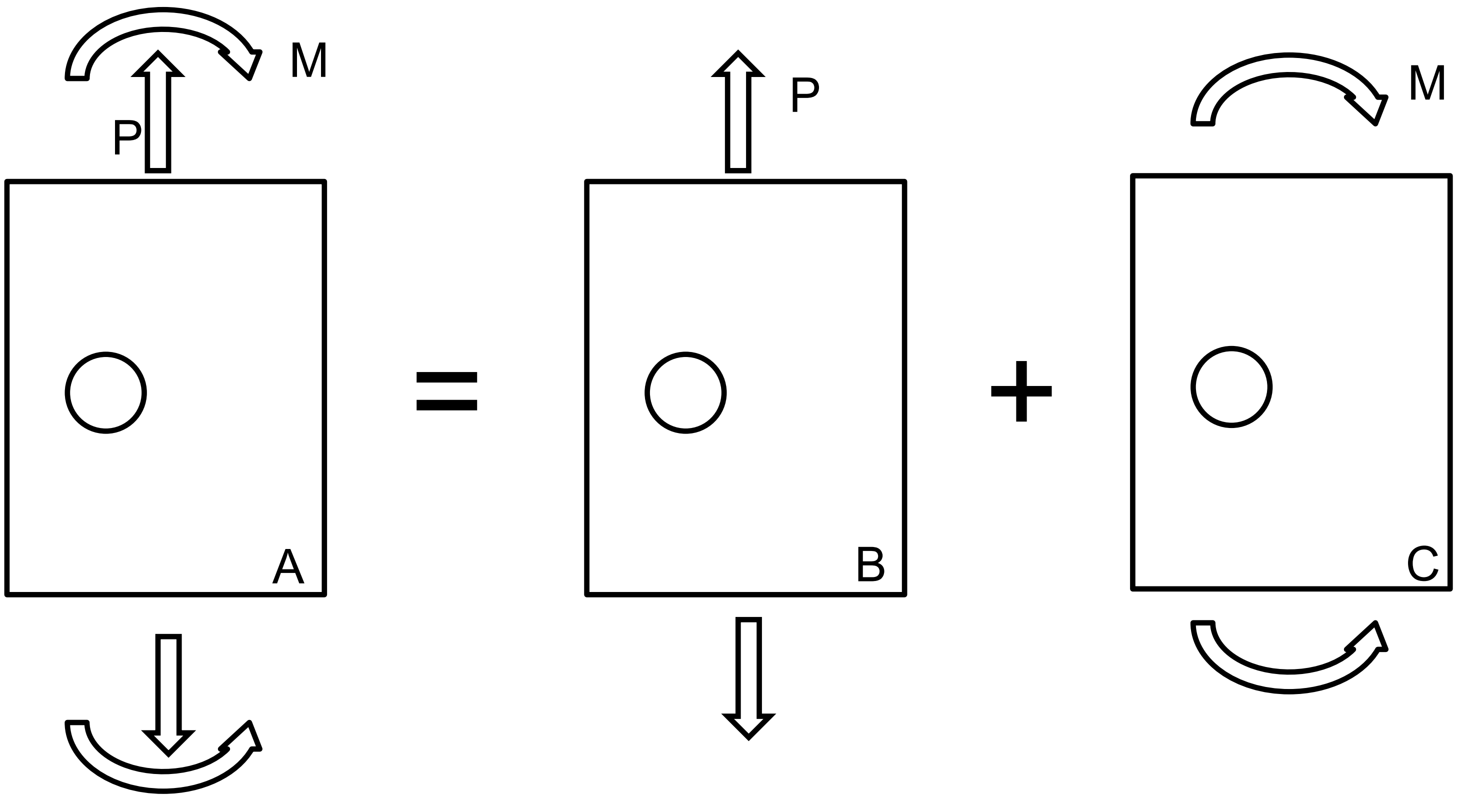
P = 2000 N

M = 20000 Nmm

EXERCISE 1 (solution)

Axial + bending problem

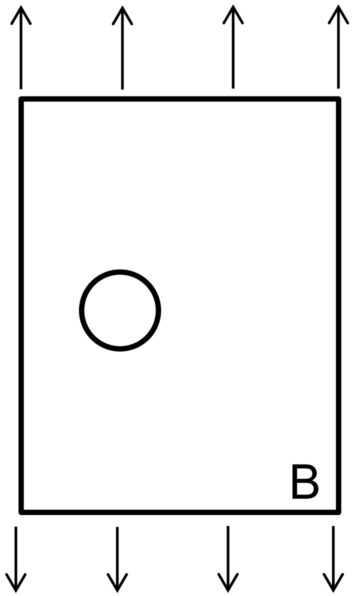
A flat plate with a hole is subjected to a combination of axial and bending loads:



$$\sigma_{max} = \sigma_{\max\,ax} + \sigma_{\max\,b}$$

WORKED EXAMPLES

EXERCISE 1 (solution)



Peterson Chart 4.3

a/c=	0.2
c/e=	0.333

	Ktg	Ktn
C1	2.99538669	2.9868688
C2	0.2354561	-2.840365
C3	0.8649895	2.413268
C4	5.64335725	N/A

Ktg	**3.1222**
Ktn	**2.5153**

$$\sigma_{ax} = P \cdot W \cdot t = 25\ MPa$$

$$\sigma_{\max\ ax} = Kt_g \cdot \sigma_{ax} = 3.1222 \cdot 25 = 78.05\ MPa$$

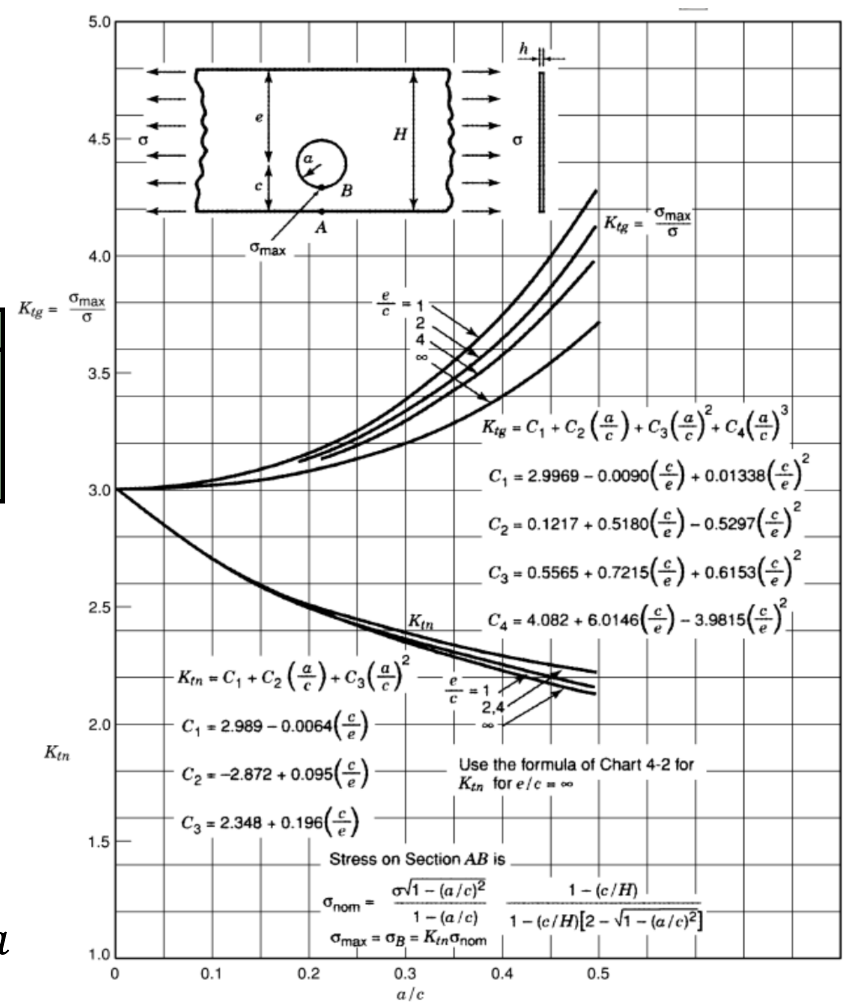
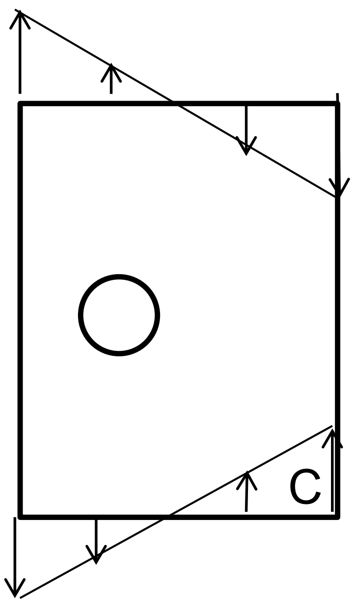


Chart 4.3 Stress concentration factors for the tension of a finite-width element having an eccentrically located circular hole (based on mathematical analysis of Sjöström 1950). $e/c = \infty$ corresponds to Chart 4.2.

EXERCISE 1 (solution)



Peterson Chart 4.80

a/c=	0.2
c/e=	0.333

	KtgB	KtgA
C1	3.03408	1.10392
C2	-4.3814	-0.160318
C3	2.43948	0.071148

KtgB	**1.8456**
KtgA	**1.0747**

$$\sigma_b = \frac{6M}{W^2 t} = 37.5\ MPa$$

$$\sigma_{\max b} = Kt_{gB} \cdot \sigma_b = 1.8456 \cdot 37.5 = 69.21\ MPa$$

356 HOLES

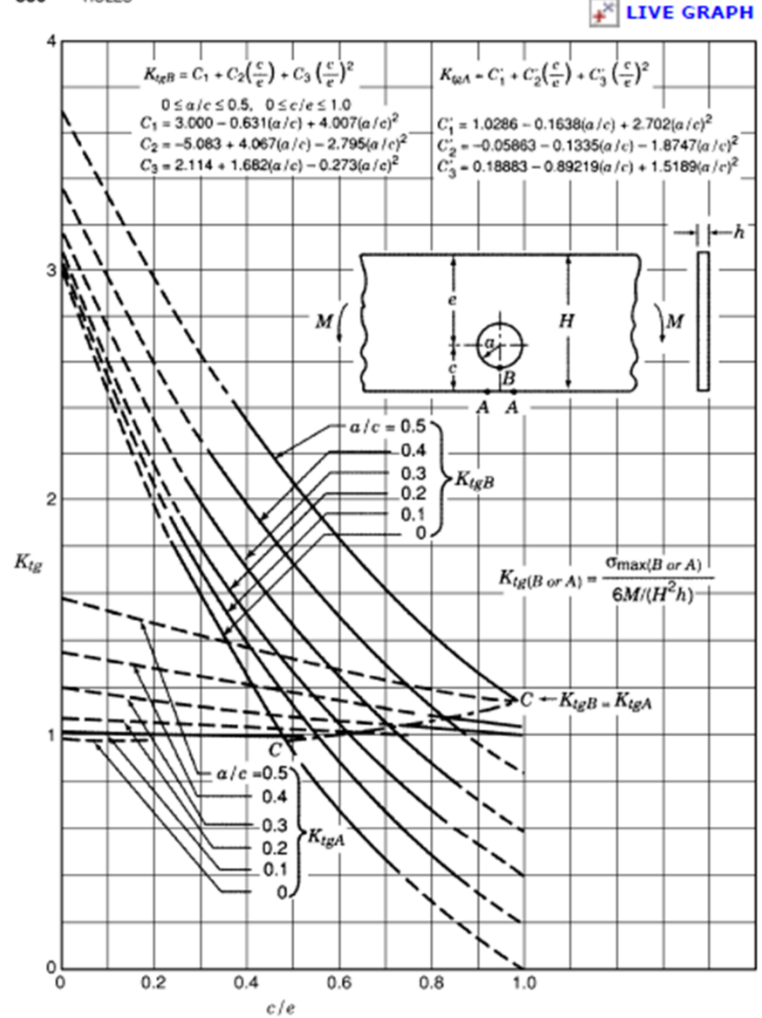
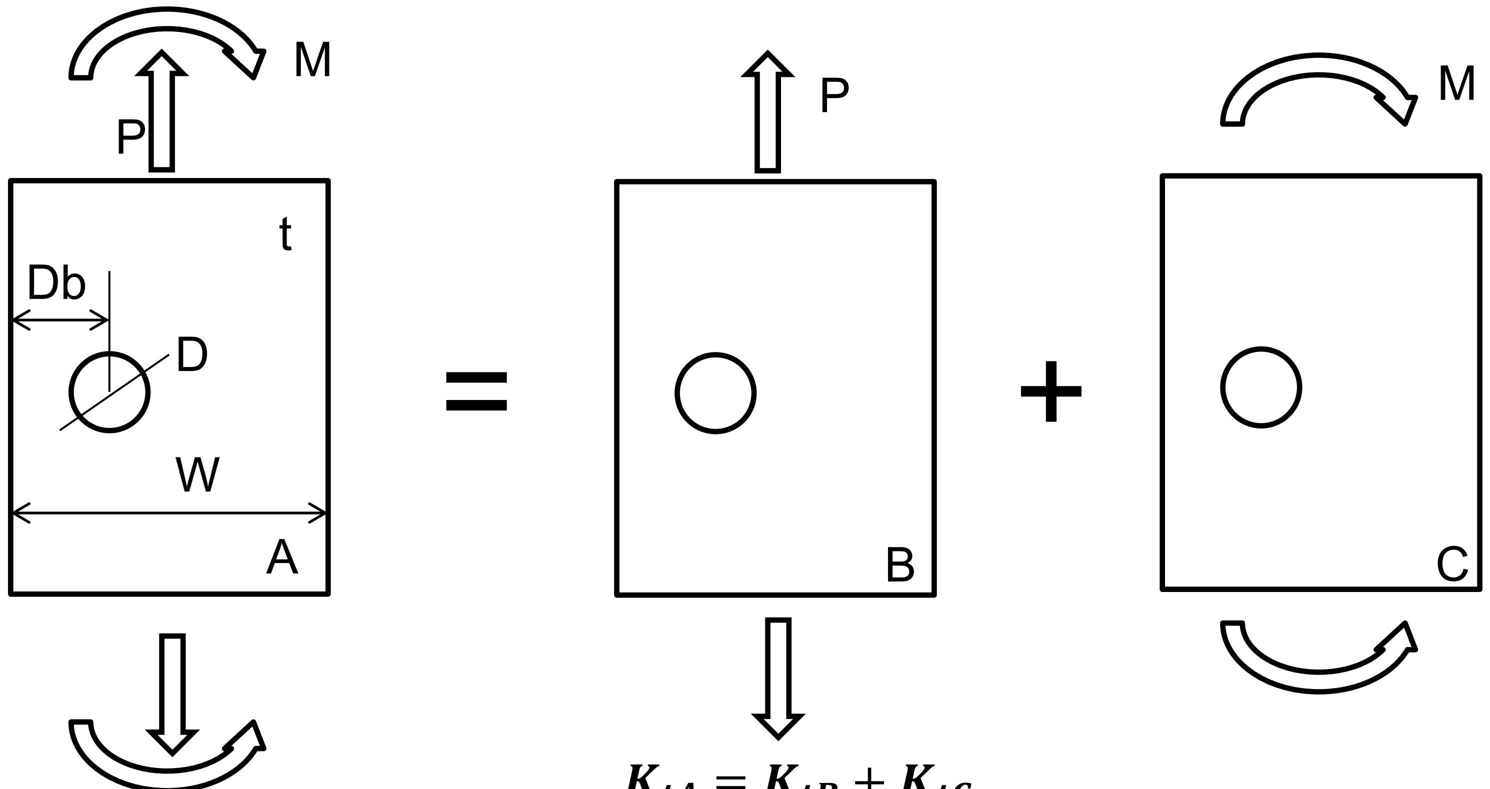


Chart 4.80 Stress concentration factors K_{tg} for bending of a thin beam with a circular hole displaced from the center line (Isida 1952).

EXERCISE 1 (solution)



$$K_{tA} = K_{tB} + K_{tC}$$

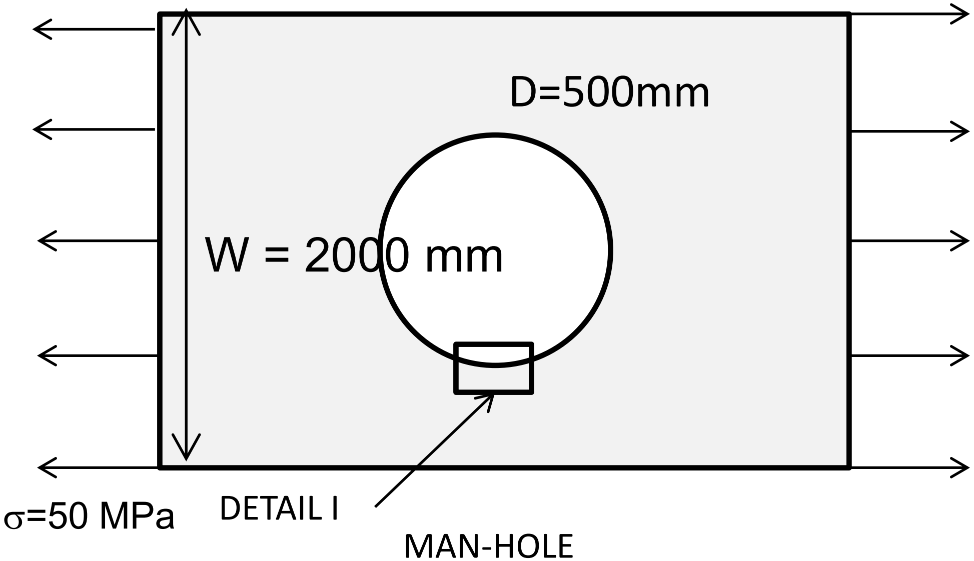
$$\sigma_g = \sigma_{g\,ax} + \sigma_{g\,b} = 25 + 37.5 = 62.5\ MPa$$

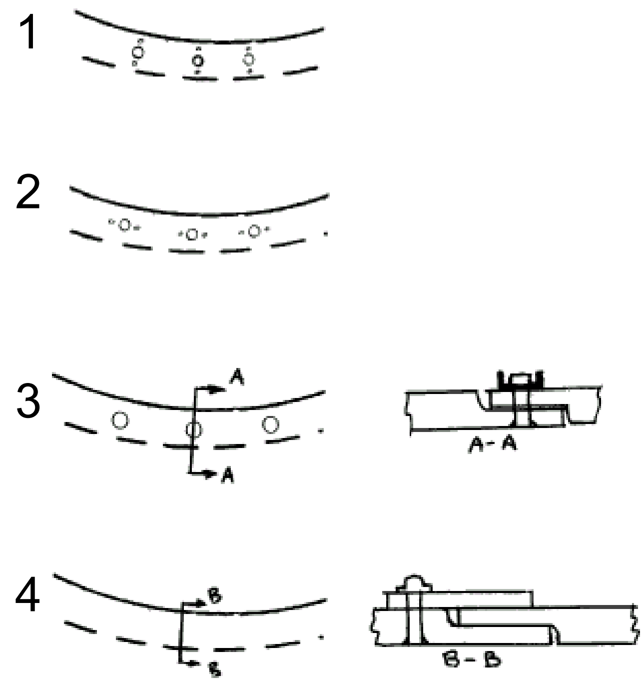
$$\sigma_{max} = \sigma_{\max ax} + \sigma_{\max b} = 147.26\ MPa$$

$$K_{tgA} = {}^{147.26}/_{62.5} = 2.3561$$

EXERCISE 2.

Calculate location of the maximum stress, the Kt and maximum stress for the 4 configurations. The bolt holes are 5 mm diameter and the nut attachment holes are 2.5 mm diameter. The bolt holes centers are located at 20 mm from the manhole edge and the nut holes centers are located at 5 mm from the bolt hole edge.





DETAIL I

EXERCISE 2. (Solution)

Configuration 4

Kt1 of manhole

Manhole in finite panel (Peterson Chart4.1)

SOLUTION: Kt maximum at manhole edge:

$Kt_{gross} = 3.2432 = {}^{\sigma_{max}}/_{\sigma}$

$Kt_{net} = 2.4324$

$$\sigma_{net} = \sigma/(1 - \frac{D}{W})$$

$$\sigma_{max} = 3.2432 \cdot 50 = 162.16\ MPa$$

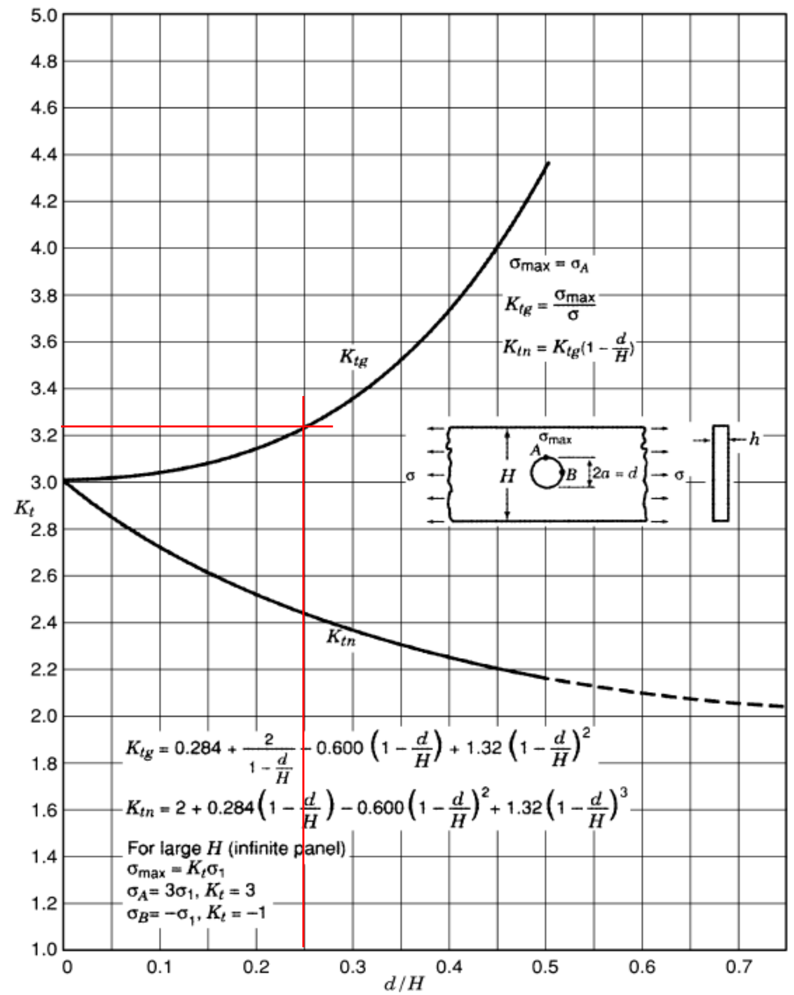


Chart 4.1 Stress concentration factors K_{tg} and K_{tn} for the tension of a finite-width thin element with a circular hole (Howland 1929–30).

EXERCISE 2. (Solution)

Configuration 3:

Kt1 of manhole in finite panel (stress at bolt hole position)

- Manhole in **infinite** panel. Stress at bolt hole position.

$K_{t1,1}$ = 2.531 (Timoshenko solution)

- Correction for **finite** panel. Compare maximum stress at manhole edge

$K_{t1,2}$ = 3.2432/3= 1.081 (Kt Chart 4.1/Kt Timoshenko)

Kt1 = $K_{t1,1} \cdot K_{t1,2}$ = 1.081·2.531 = 2.736

$\sigma 1$ = 2.736 ·50 = 136.8 Mpa (stress at bolt hole position)

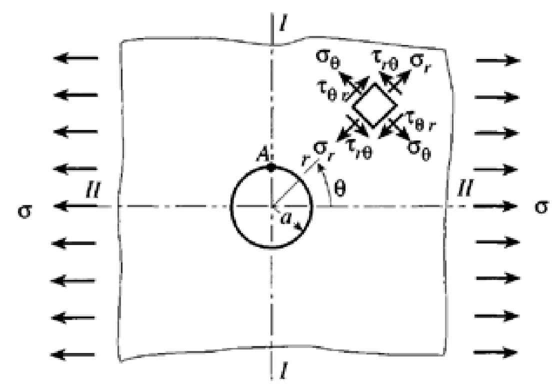
Kt2 of bolt hole as isolated hole in infinite panel

Kt2 = K_t (Bolt) = 3 = $\sigma_{max}/\sigma 1$

Stress distribution around a hole (Timoshenko)

stress gross	a (radius)	r	teta	teta (radians)
50	250	270	90	1.570795

		stresses ratio	stress (total)
sigma r	9.2	2.531	126.893
sigma teta	126.6		
tau			



SOLUTION: Kt maximum at bolt hole edge:

Kt_{gross}= 2.736 · 3 = 8.208 = σ_{max}/σ

σ_{max}= 8.208 · 50 = 410.4 MPa (maximum stress at bolt hole edge)

EXERCISE 2. (Solution)

Configuration 2

<u>Kt1 of manhole in finite panel (stress at bolt hole position)</u>

$Kt1 = K_{t1,1} \cdot K_{t1,2} = 1.081 \cdot 2.531 = 2.736$

$\sigma 1 = 2.736 \cdot 50 = 136.8$ Mpa (stress at bolt hole position)

<u>Kt2 of two holes (bolt + nut) of unequal diameter aligned in an infinite panel</u> (Peterson Chart 4.27)

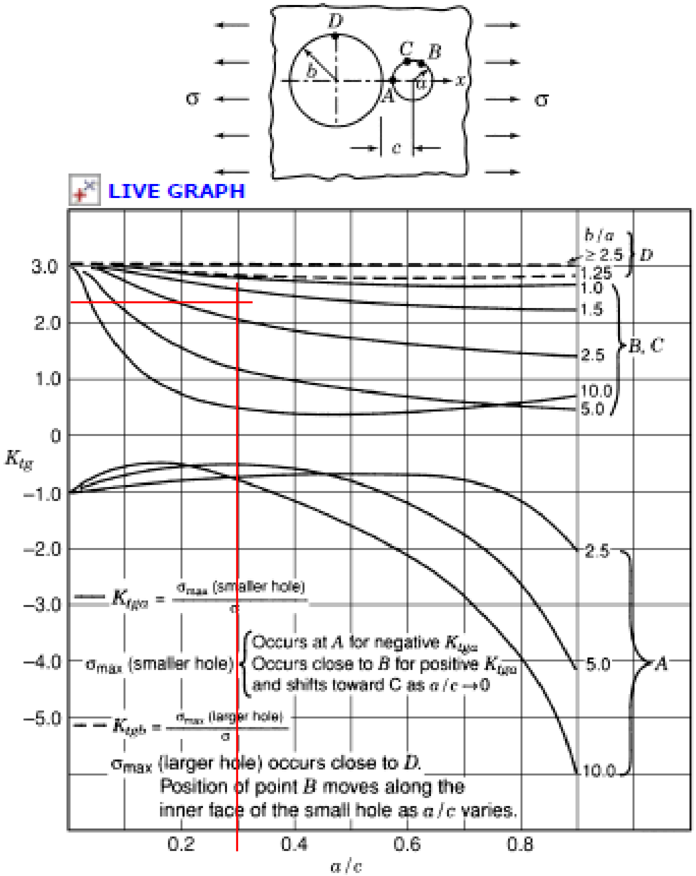
$Kt2.1 = K_{tD}$ (Bolt hole) $= 3 = {}^{\sigma max}/_{\sigma 1}$

$Kt2.2 = K_{tB} = K_{tC}$ (Nut hole) $= 2.4 = {}^{\sigma max}/_{\sigma 1}$

SOLUTION: Kt maximum at bolt hole edge:

$Kt_{gross} = 2.736 \cdot 3 = 8.208 = {}^{\sigma max}/_{\sigma}$

$\sigma_{max} = 8.208 \cdot 50 = 410.4$ MPa (maximum stress at bolt hole edge)



EXERCISE 2. (Solution)

Configuration 1

Kt1 of manhole in finite panel (stress at bolt hole position)

$Kt1 = K_{t1,1} \cdot K_{t1,2} = 1.081 \cdot 2.688 = 2.9057$

$\sigma 1 = 2.9057 \cdot 50 = 145.3$ Mpa (stress at nut hole position)

Kt2 of two holes (bolt + nut) of unequal diameter aligned in an infinite panel (Peterson Chart 4.26)

$Kt2 = K_{tD}$ (Nut hole) $= 3.4 = {}^{\sigma_{max}}/_{\sigma 1}$

SOLUTION: Kt at nut hole edge:

$Kt_{gross} = 3.4 \times 2.9057 = 9.879 = {}^{\sigma_{max}}/_{\sigma}$

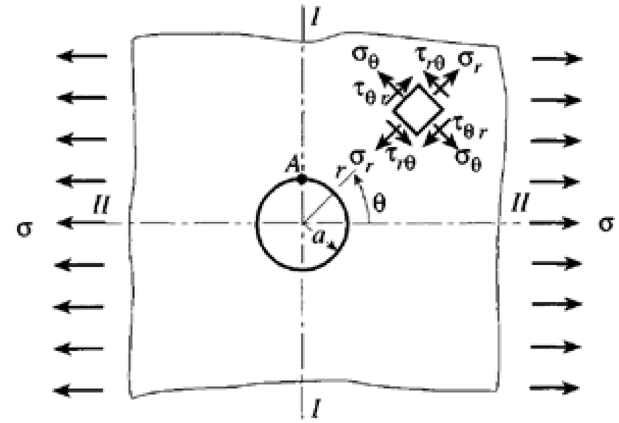
$\sigma_{max} = 9.879 \cdot 50 = 494$ MPa (maximum stress at nut hole edge)

Stress distribution around a hole (Timoshenko)

stress gross	a (radius)	r	teta	teta (radians)
50	250	262.5	90	1.570795

sigma r	6.3
sigma teta	134.4
tau	

stresses ratio	stress (total)
2.688	134.527



EXERCISE 2. (Solution)

a=1.25 mm

b=2.5 mm

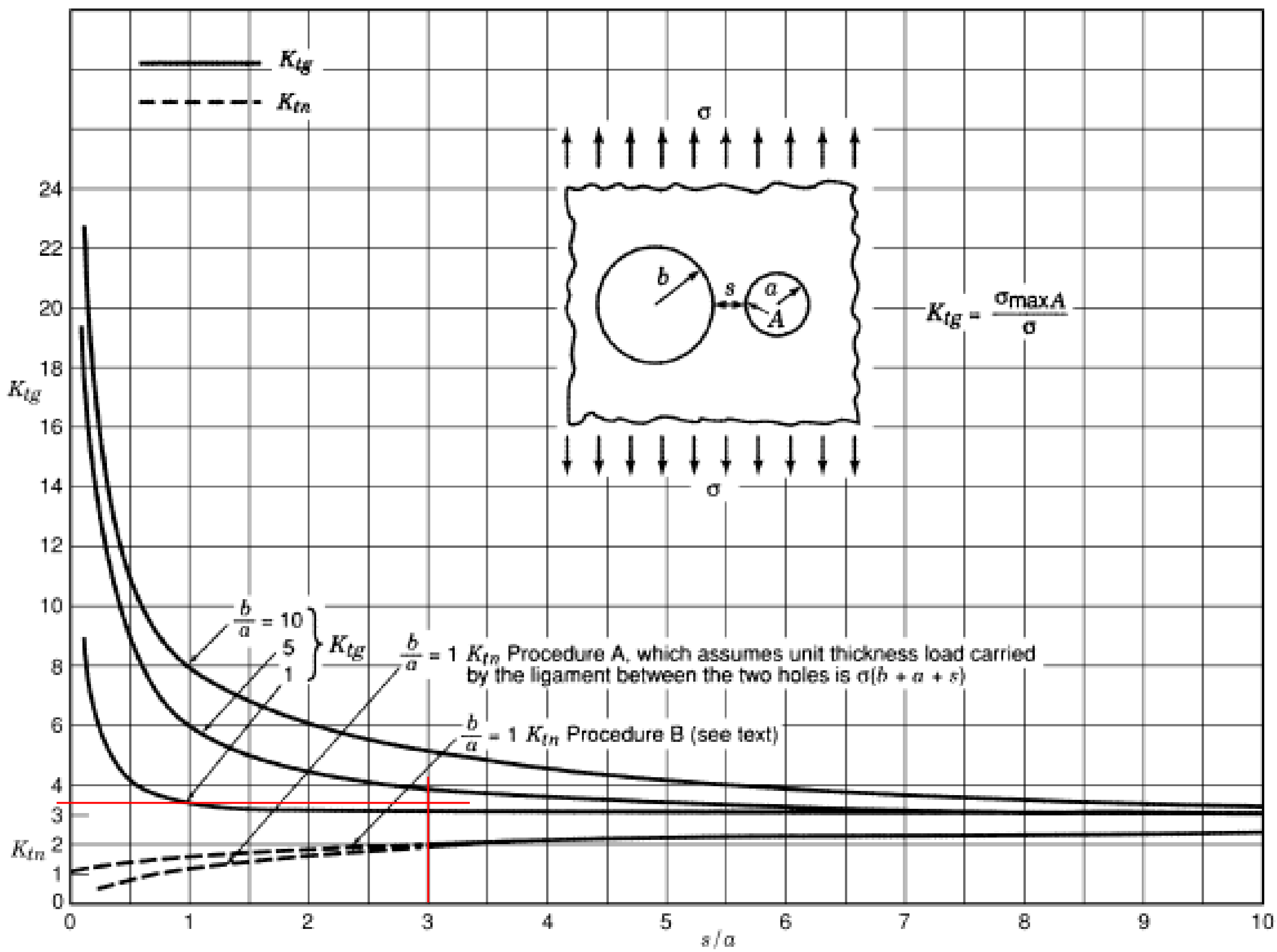
s= 3.75 mm

b/a = 2

s/a = 3

Ktg = 3.4

LIVE GRAPH



EXERCISE 3

The joint in the figure is made by means of 2 countersunk rivets. Identify the critical location for fatigue. Calculate the Kt_{net} at the critical location. Apply the following additional factors to the Kt appropriately.

t= 2 mm

W = 20 mm

H= 50 mm

D = 4 mm

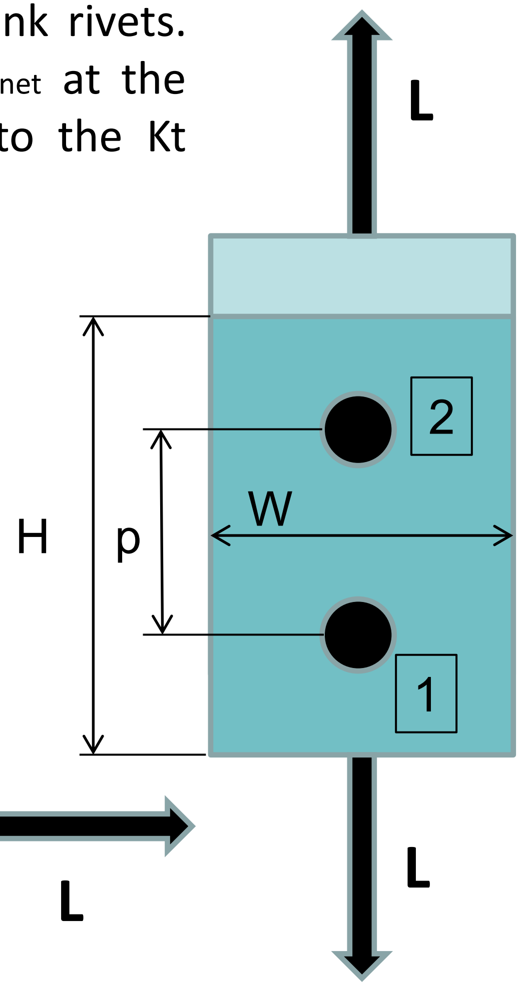
L= 5000 N

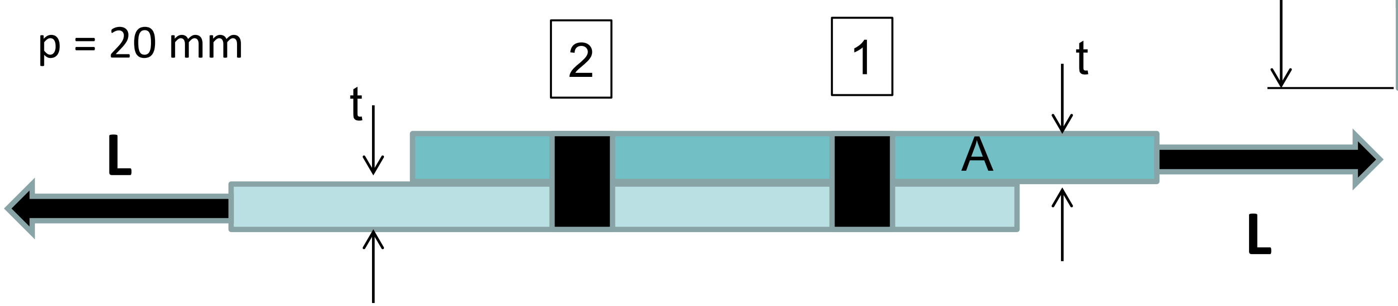
p = 20 mm

Kt_{pb} = 1.15 (pin bending)

Kt_{th} = 1.0235 (thickness)

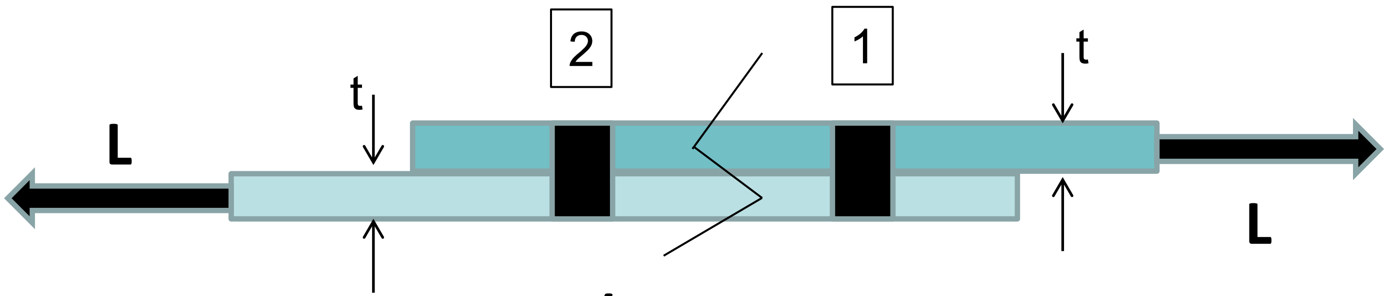
Kt_{csk}=1.2297 (countersunk)

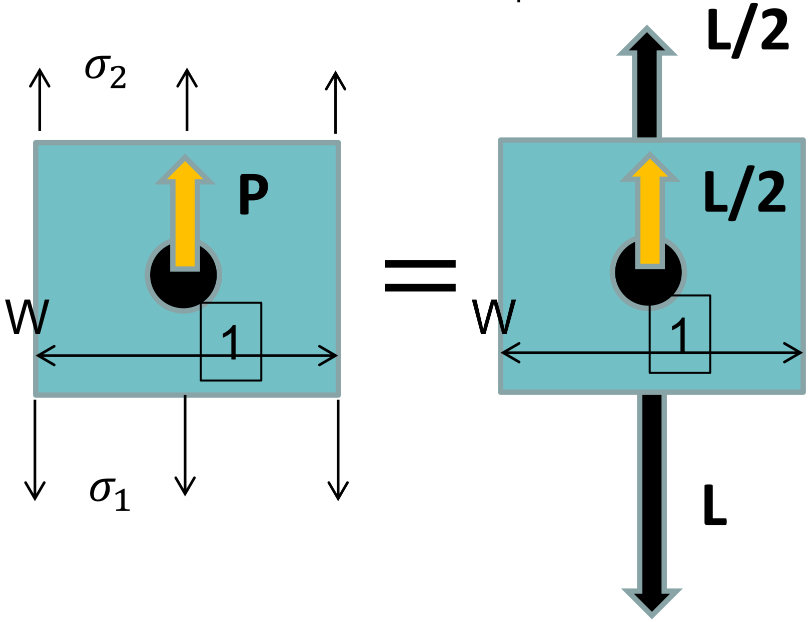


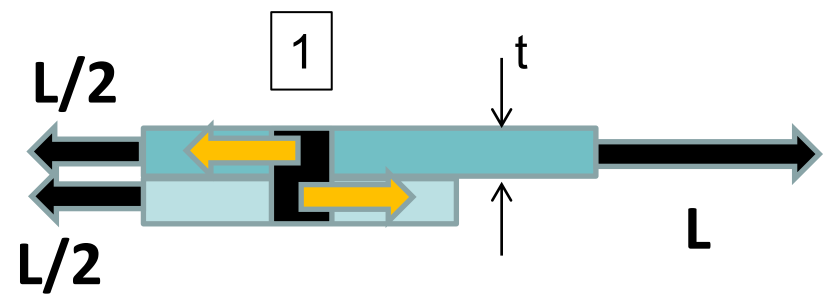


EXERCISE 3. (Solution)

Step1: Isolate the holes and calculate the forces equilibrium and the far field stresses:







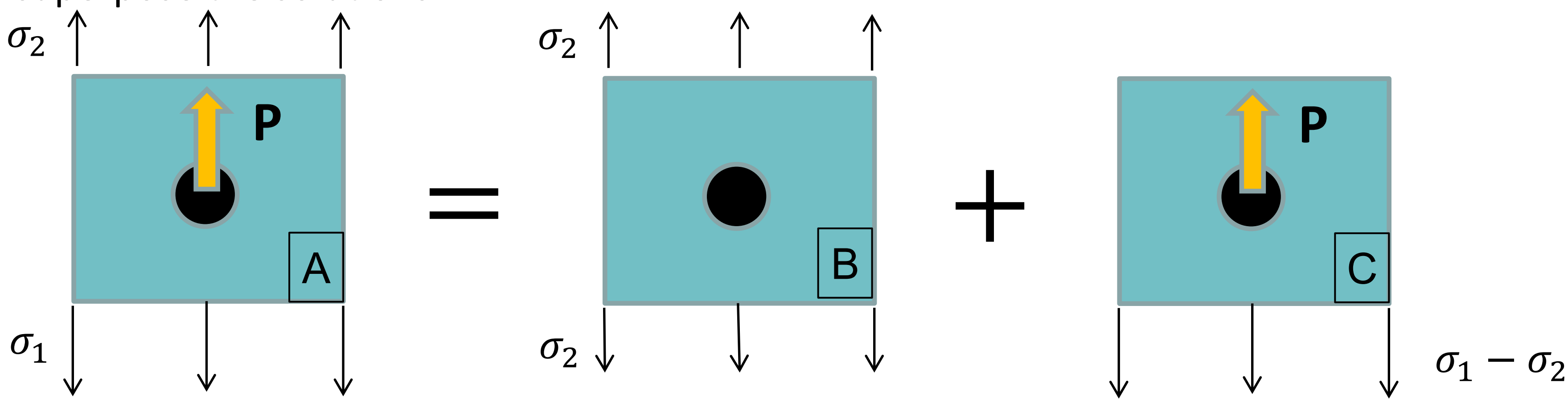
$$\sigma_1 = \frac{L}{Wt} = 125\text{MPa}$$

$$\sigma_2 = \frac{L/2}{Wt} = 62.5\text{MPa}$$

$$\text{P} = \frac{\text{L}}{2} = 2500\text{N} = (\sigma_1 - \sigma_2) \times Wt$$

EXERCISE 3. (Solution)

Step2: Decompose the configuration in more simple configurations with known solutions for Kt. Apply the principle of superposition. Refer Kt to the same stress to superpose the solutions:



$$\sigma_{\max A} = \sigma_{\max B} + \sigma_{\max C}$$

$$Kt_{gA,1} = Kt_{gB,1} + Kt_{gC,1}$$

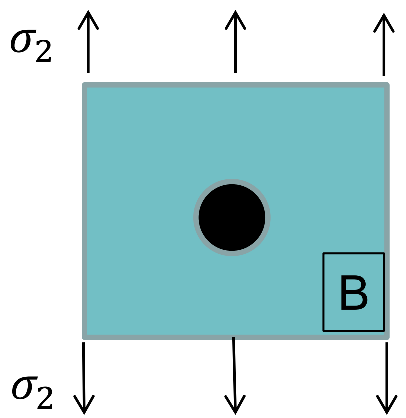
$$\sigma_{gross} = \sigma_1$$

$$\sigma_{net} = \sigma_1 \cdot SAF_{net} = \sigma_1 \cdot \frac{1}{\left(1 - {}^{D}/_{W}\right)} = 125 \cdot 1.25 = 156.25\ MPa$$

EXERCISE 3. (Solution)

Step3: Calculate the Kt applying the principle of superposition for simple configurations with Peterson, ESDU or other published sources.

Kt_B: Peterson Chart4.1



d/H=	0.2

Ktg	**3.1488**
Ktn	**2.5190**

$Ktg_{B,2} = 3.1488 = \sigma_{maxB}/\sigma_2$

$Ktg_{B,1} = \sigma_{maxB}/\sigma_1 = \sigma_{maxB}/\sigma_2 \cdot \sigma_2/\sigma_1 = 3.1488 \cdot 0.5 = 1.5744$

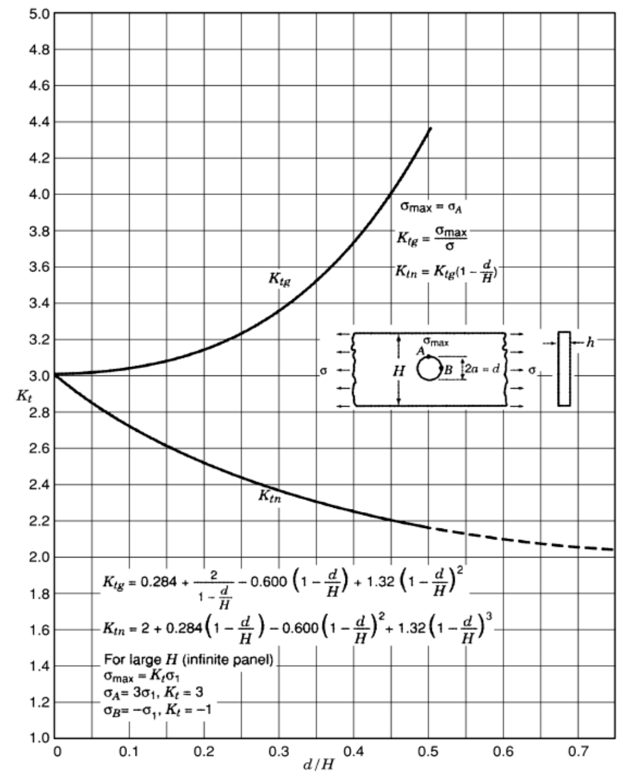
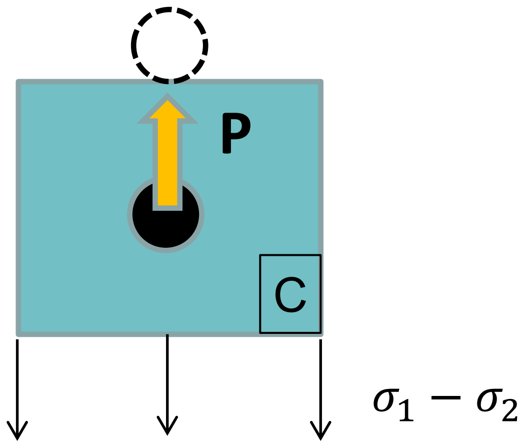


Chart 4.1 Stress concentration factors K_{tg} and K_{tn} for the tension of a finite-width thin element with a circular hole (Howland 1929–30).

EXERCISE 3. (Solution)

Step3 (cont'd):

Ktc: ESDU 81006 (squared lugs)



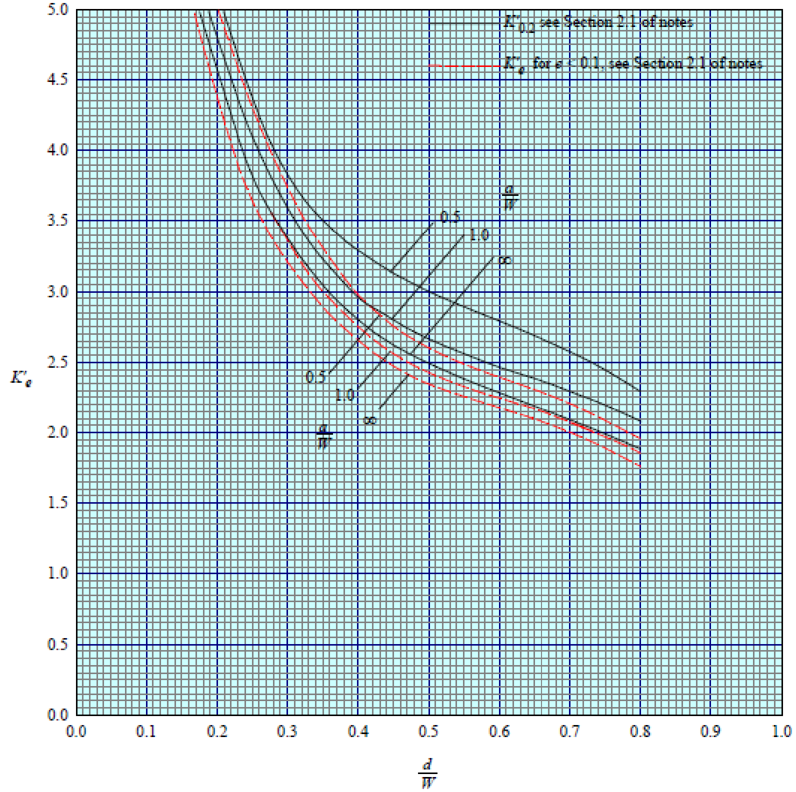
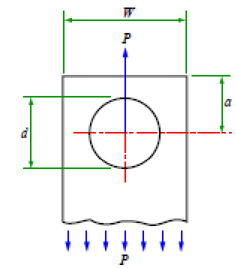


FIGURE 1 SQUARE-ENDED LUGS



d/W=0.2

a=20-2= 18 mm ; a/W = 0.9

$Ktn_C = K'_{0.2} = 4.9 = \sigma_{maxC} / \frac{P}{(W-D)t}$

$Ktg_{C,1} = {}^{\sigma_{maxC}}/_{\sigma_1} = 4.9 \cdot \frac{P}{\sigma_1 (W-D)t} = 4.9 \cdot 0.625 = 3.0625$

EXERCISE 3. (Solution)

<u>Kt with hole effects:</u>

Kt_{pb} = 1.15 (pin bending)

Kt_{th} = 1.0235 (thickness)

Kt_{csk}=1.2297 (countersunk)

(Contact zone)
$Kt_{gA,1} = (Kt_{gB,1} + Kt_{pb} \cdot Kt_{gC,1}) = (1.5744 + 1.15 \cdot 3.0625) = 5.096$
$Kt_{net,A} = Kt_{gA,1}/SAF_{net} = \mathbf{4.08}$

(Cylindrical zone)
$Kt_{gA,1} = Kt_{th} \cdot (Kt_{gB,1} + Kt_{gC,1}) = 1.0235 \cdot (1.5744 + 3.0625) = 4.746$
$Kt_{net,A} = Kt_{gA,1}/SAF_{net} = \mathbf{3.8}$

(Countersunk zone)
$Kt_{gA,1} = Kt_{csk} \cdot (Kt_{gB,1} + Kt_{gC,1}) = 1.2297 \cdot (1.5744 + 3.0625) = 5.702$
$Kt_{net,A} = Kt_{gA,1}/SAF_{net} = \mathbf{4.56}$

EXERCISE 4

CALCULATE THE Kt_{net} AND Kt_{gross} OF THE FOLLOWING LUG. USE ESDU81006

Location: Male lug at typical wing to fuselage attachment.

There are **two lugs** which are identical, each one with half load. If one lug fails the other one can carry the total load.

Each lug thickness is **12.75 mm** and the material **7075-T7351.** The pin material is **steel.**

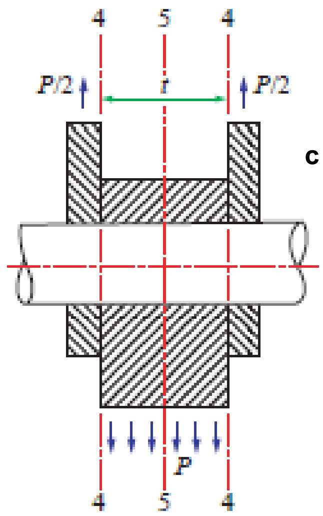
Lug hole diameter is: $\phi_{lug} = \mathbf{54^{+0.10}_{+0.01}}$ **mm**

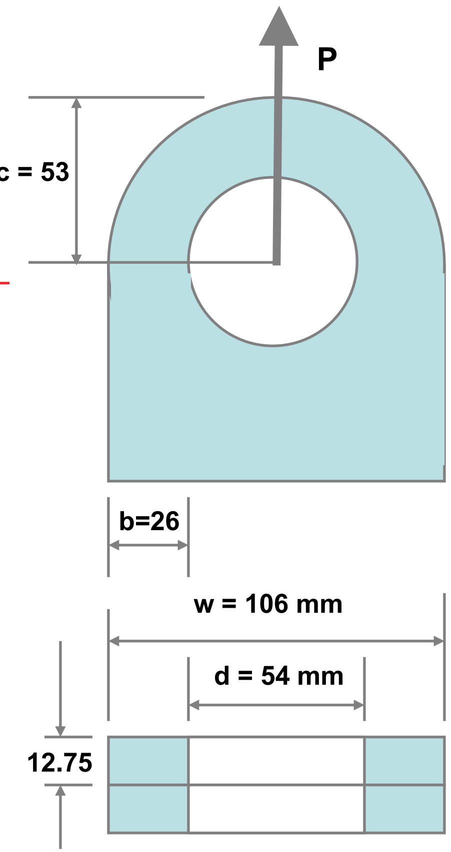
Pin diameter is: $\phi_{pin} = \mathbf{54^{-0.08}_{-0.01}}$ **mm**

$\sigma_{gross} = P/(W \cdot 2t)$

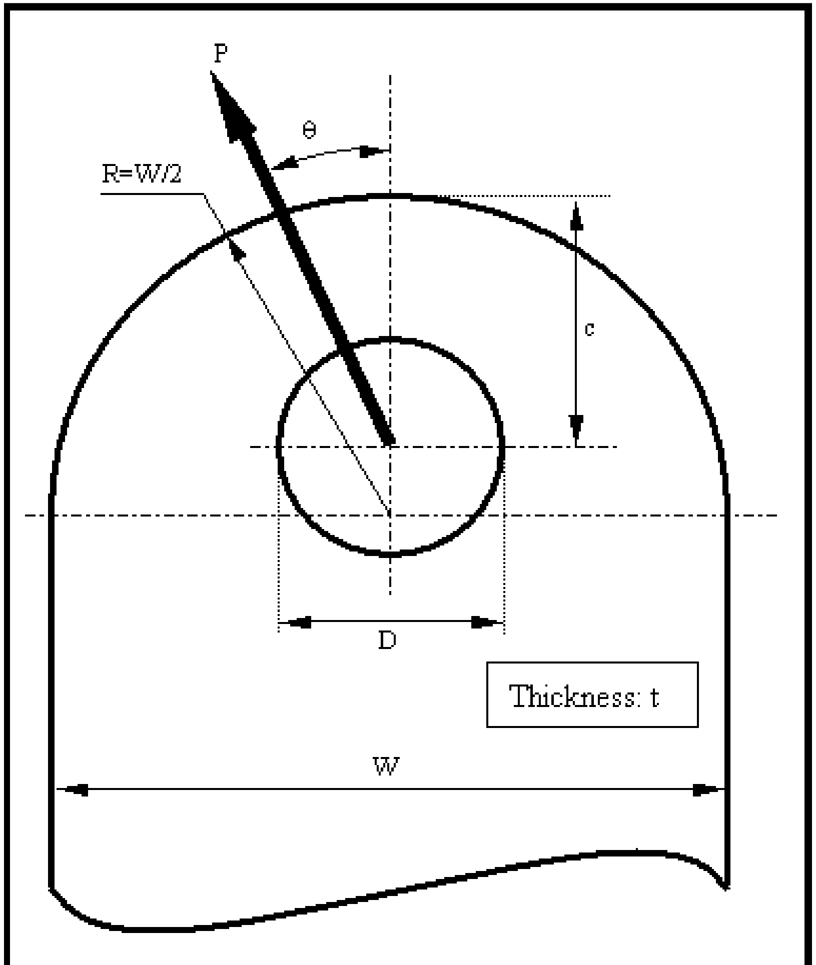
$SAF = \sigma_{gross}/P = 1/(W \cdot 2t) = 3.7 \cdot 10^{-4}$

$$e = \frac{(0.10 + 0.08) \cdot 100}{54} = 0.333$$





EXERCISE 4 (solution ESDU81006)



INPUT DATA			
Geometrical Data			
R (mm)	53	c (mm)	53
D (mm)	54	t (mm)	25.5
W (mm)	106	θ (°)	0
Load (1: Single Shear; 2: Double Shear)			2
Materials (1: Aluminium; 2: Titanium; 3:Steel)			
Bolt Material	3	Lug Material	1
Shim and Clearance (No effect with null values)			
t_{shim} (mm)	0	e (%)	0.333

RESULTS	
Basic $K'_{0.2}$	3.065
Pin Bending Effect ($K_{t,pb}$)	1.020
Shim Effect ($K_{t,shim}$)	NO
Angle Effect ($K_{t,\theta}$)	1.000
Clearance Effect (K'_{100})	4.068
Clearance Factor (η_e)	0.117
$K_{t,net}$	3.253
$K_{t,gross}$	6.632
$K_{t,brg}$	3.378

REMARKS

$$K'_e = K'_{0.2} + \eta_e(K'_{100} - K'_{0.2})$$

$$f_{max} = K'_e f_{ref}$$

$$f_{ref} = \frac{P}{(W-d)t}$$

L04. References

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